

AKADEMIA GÓRNICZO-HUTNICZA
IM. STANISŁAWA STASZICA W KRAKOWIE

AGH UNIVERSITY OF SCIENCE
AND TECHNOLOGY

Static Structural Analysis in Marc Mentat

Materiały i Struktury Inteligentne (2018) – przygotował: Jakub Bryła
opieka merytoryczna: dr inż. Adam Martowicz

Agenda:

-> Problem description

-> Analytical solution

-> 1D FEM simulation

-> 3D FEM simulation

-> 3D FEM – result improvment methods

-> Results comparison

www.agh.edu.pl

Problem description

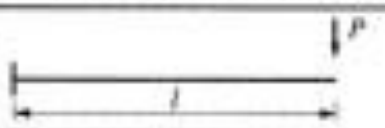
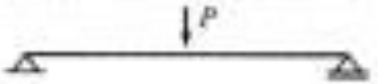
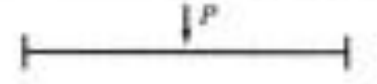
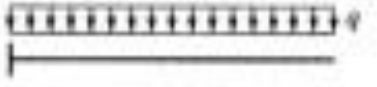
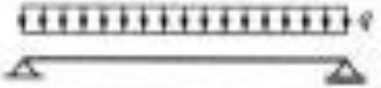
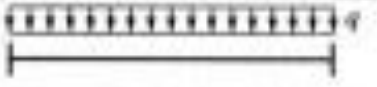
The goal is to find maximum deflection of a beam.

Geometry data: The beam has rectangular cross section with side dimensions 4 and 8mm. The length of the item is 50mm. The beam is made of the steel.

Boundary condition: The beam is fixed on one side, while on the other side a 650N force is applied at center of the cross section, along shorter side.

www.agh.edu.pl

Analytical solution

Schemat statyczny	Maksymalne ugięcie
	$\delta = \frac{Pl^3}{3EJ}(1+3k)$
	$\delta = \frac{Pl^3}{48EJ}(1+12k)$
	$\delta = \frac{Pl^3}{107,5EJ}(1+27,7k)$
	$\delta = \frac{Pl^3}{192EJ}(1+48k)$
	$\delta = \frac{ql^4}{8EJ}(1+4k)$
	$\delta = \frac{5ql^4}{384EJ}(1+9,6k)$
	$\delta = \frac{ql^4}{185EJ}(1+22,6k)$
	$\delta = \frac{ql^4}{384EJ}(1+48k)$

$k = \frac{EJ}{5,l^2}$

$$\delta = \frac{P\,l^3}{3\,E\,J}$$

$$E = 210\ GPa = 210\ \cdot 10^9\ Pa = 2.1 \cdot 10^{11}\ \frac{N}{m^2}$$

$$= 2.1 \cdot 10^{11}\ \frac{N}{10^6 mm^2} = 2.1 \cdot 10^5\ \frac{N}{mm^2}$$

$$J = \frac{b \cdot h^3}{12} = \frac{8mm\ \cdot (4mm)^3}{12} = \frac{128}{3}\ mm^4$$

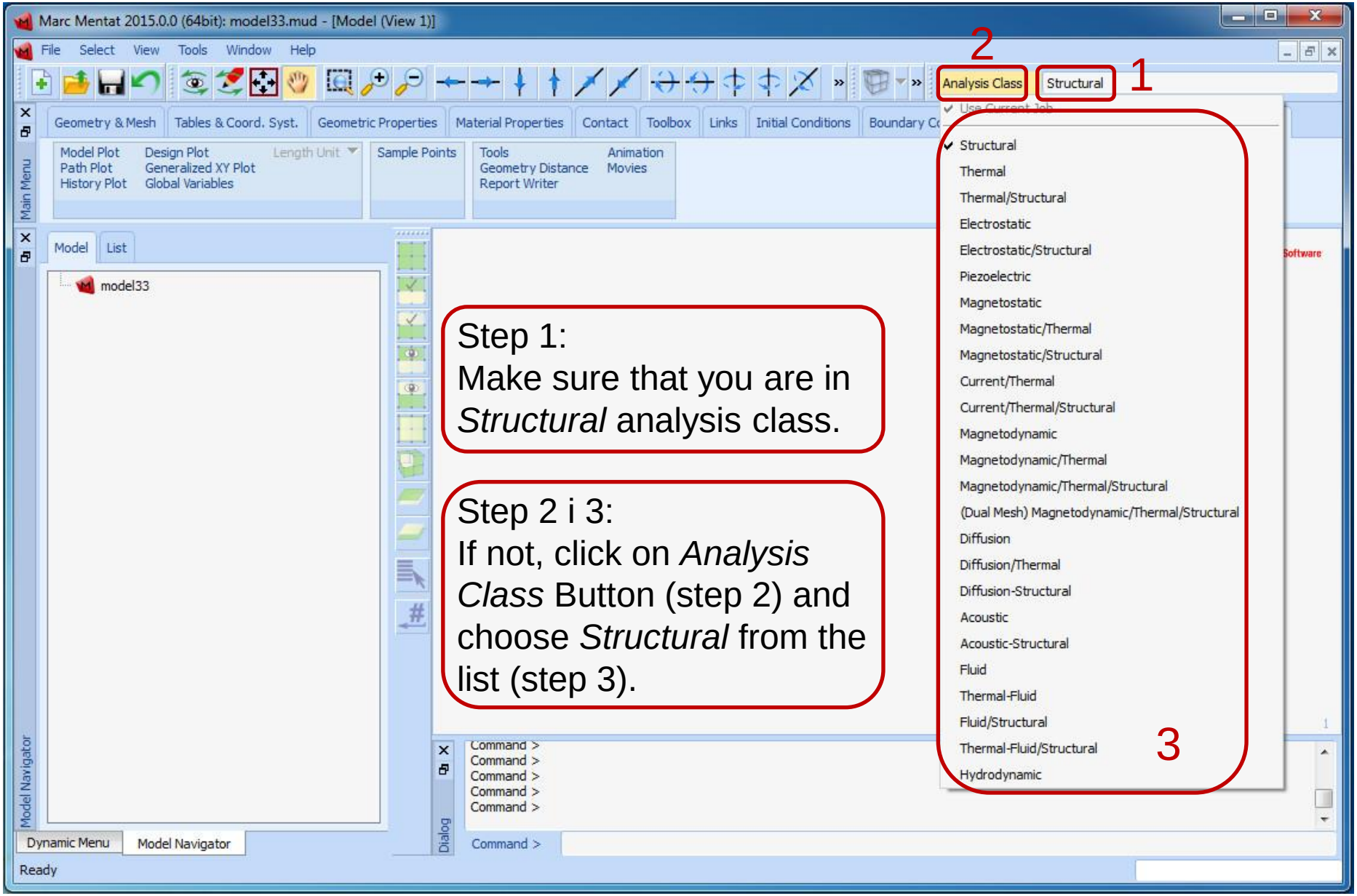
$$\delta = \frac{650N\ \cdot (50mm)^3}{3\ \cdot 2.1 \cdot 10^5\ \frac{N}{mm^2}\ \cdot \frac{128}{3}\ mm^4}$$

$$\delta = 3.0227mm$$

Source: http://chodor-projekt.net/encyclopedia/stalowe-dzwigary-kratowe/

www.agh.edu.pl

1D FEM solution



1D FEM solution

Create model in one of the known you method. In this case we use the second method described in *Geometry & Mesh* tutorial.

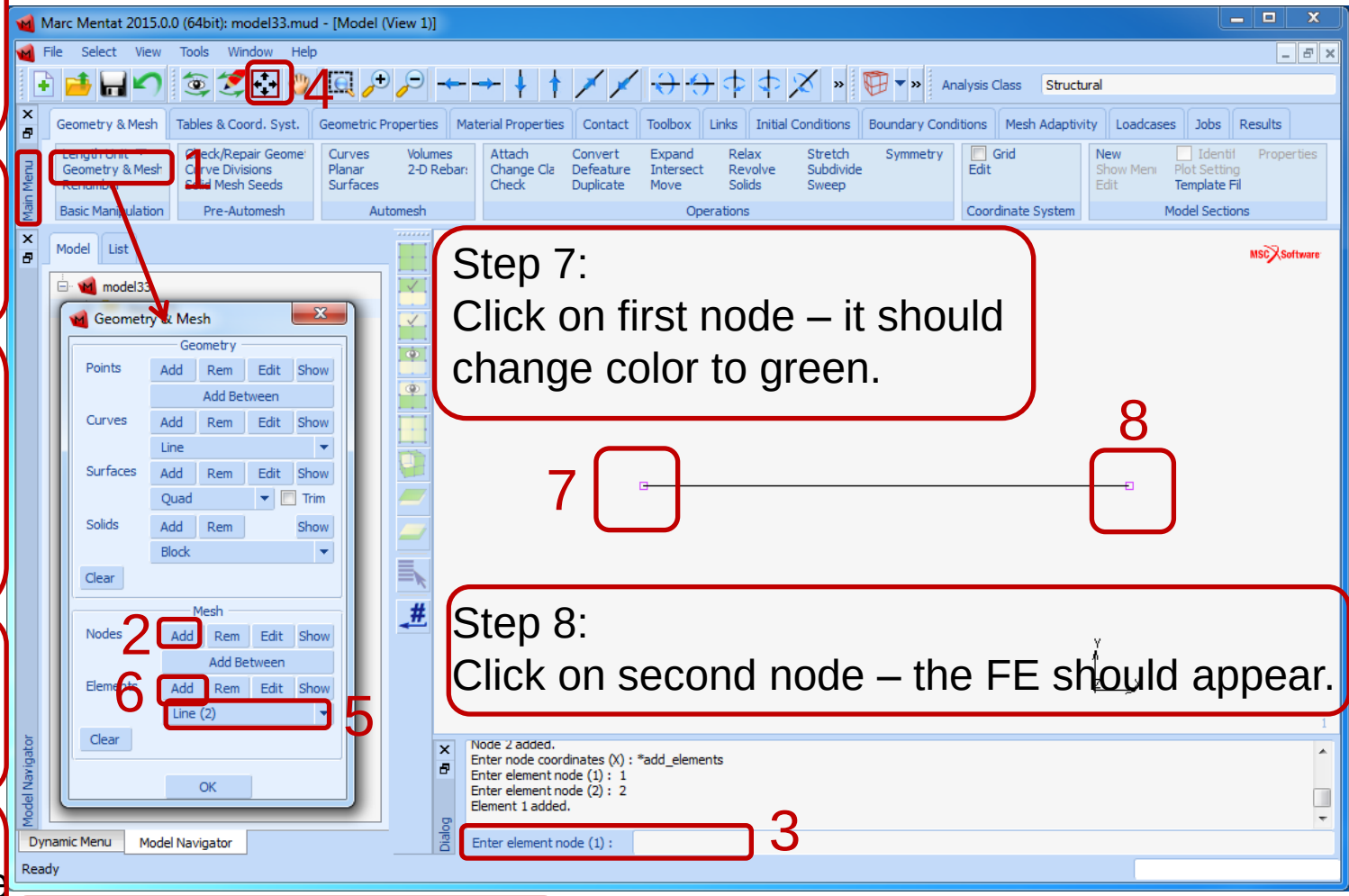
Step 1:
Open *Geometry & Mesh* Menu from *Geometry & Mesh* Tab located in Main Manu Tab.

Step 2:
Click *Add* Button to add nodes.

Step 3:
Input coordinates data to *Command Window*:
-> 0 0 0 (Enter)
-> 50 0 0 (Enter)

Step 4:
Click *Fill view* Button to see all nodes in Model View.

Step 5:
Make sure that you will create 1D FE with 2nodes: option *Line(2)*



Step 6:
Click Add Button to add FE.

www.agh.edu.pl

1D FEM solution

Step 1:
Open Subdivide Menu from Operations Tab.

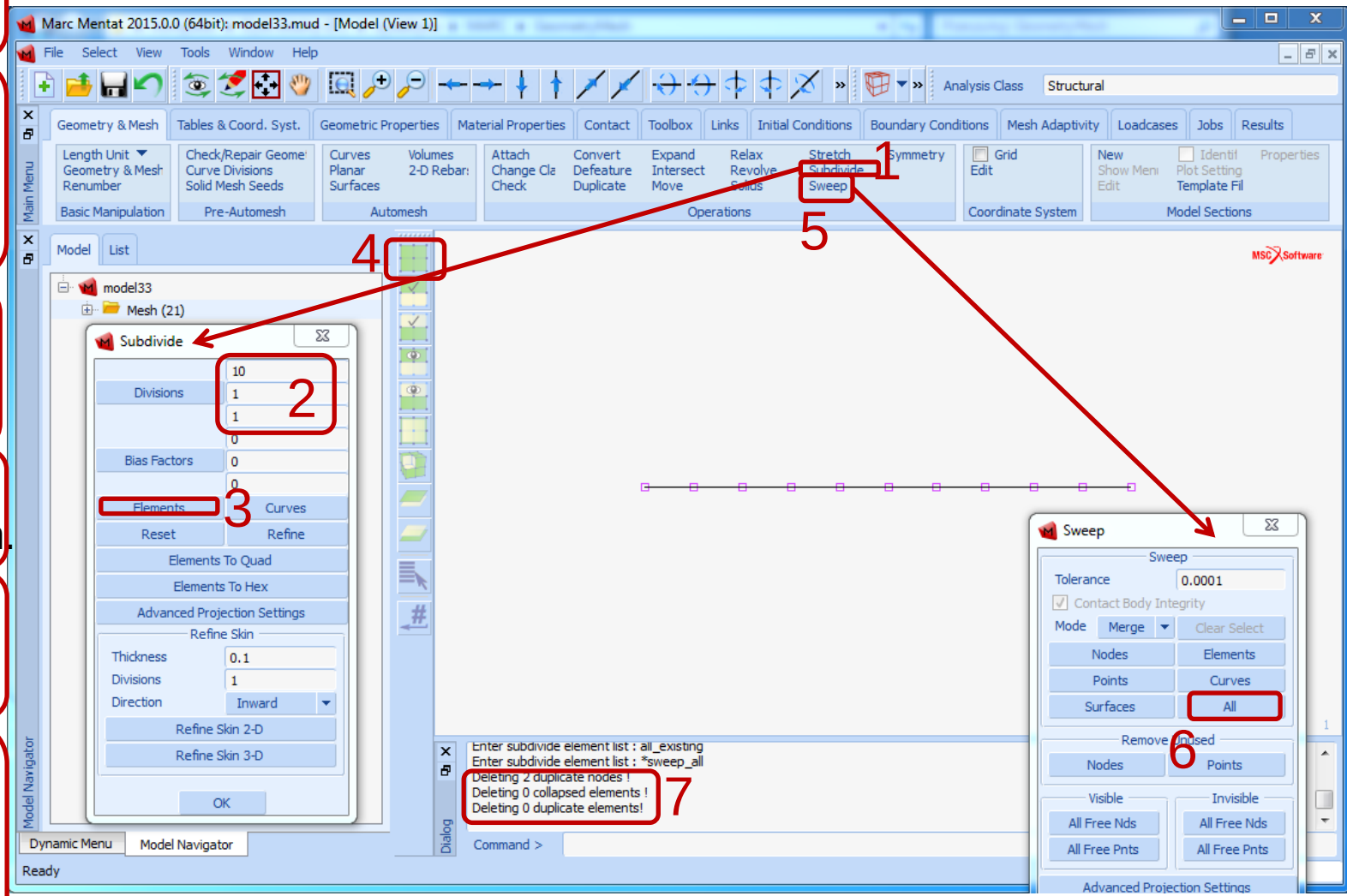
Step 2:
Input *Divisions* data respectively along each axis.

Step 3:
Click *Elements* Button to divide FE.

Step 4:
Click *All Existing* Button.

Step 5:
Open *Sweep* Menu from *Operations* Tab.

Step 6:
Click *All* Button to delete duplicated items. Note that 2 nodes were deleted – see 7 .



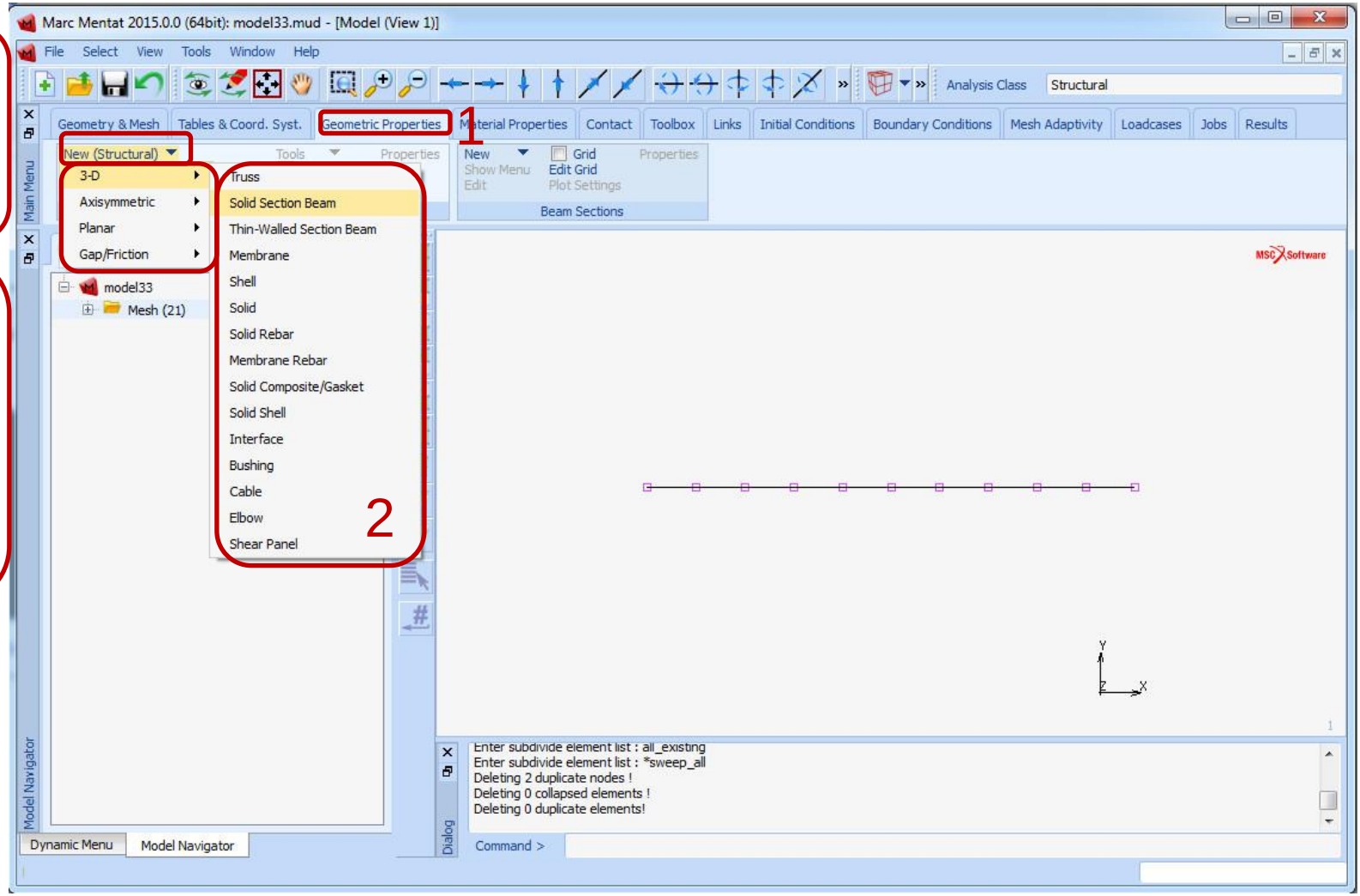
www.agh.edu.pl

1D FEM solution

All 1D and 2D FE types need to define Geometric Properties!

Step 1:
Go to *Geometric Properties* Tab from *Main Menu*.

Step 2:
Click on *New (Structural)* list and choose: *3-D* option and next *Solid Section Beam* option.



1D FEM solution

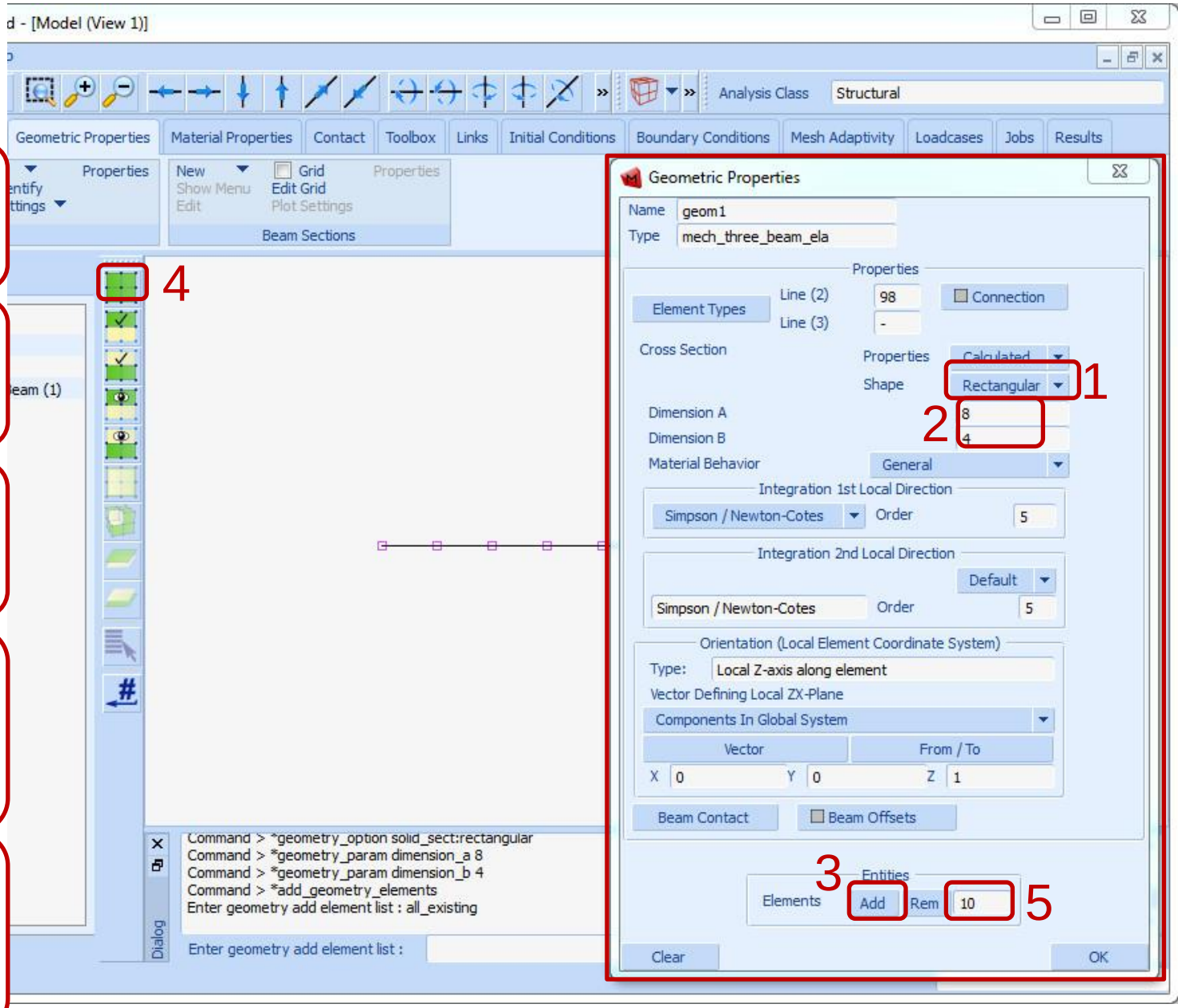
Action from previous slide opens *Geometric Properties* Menu.

Step 1:
Change shape of the Cross Section to *Rectangular*.

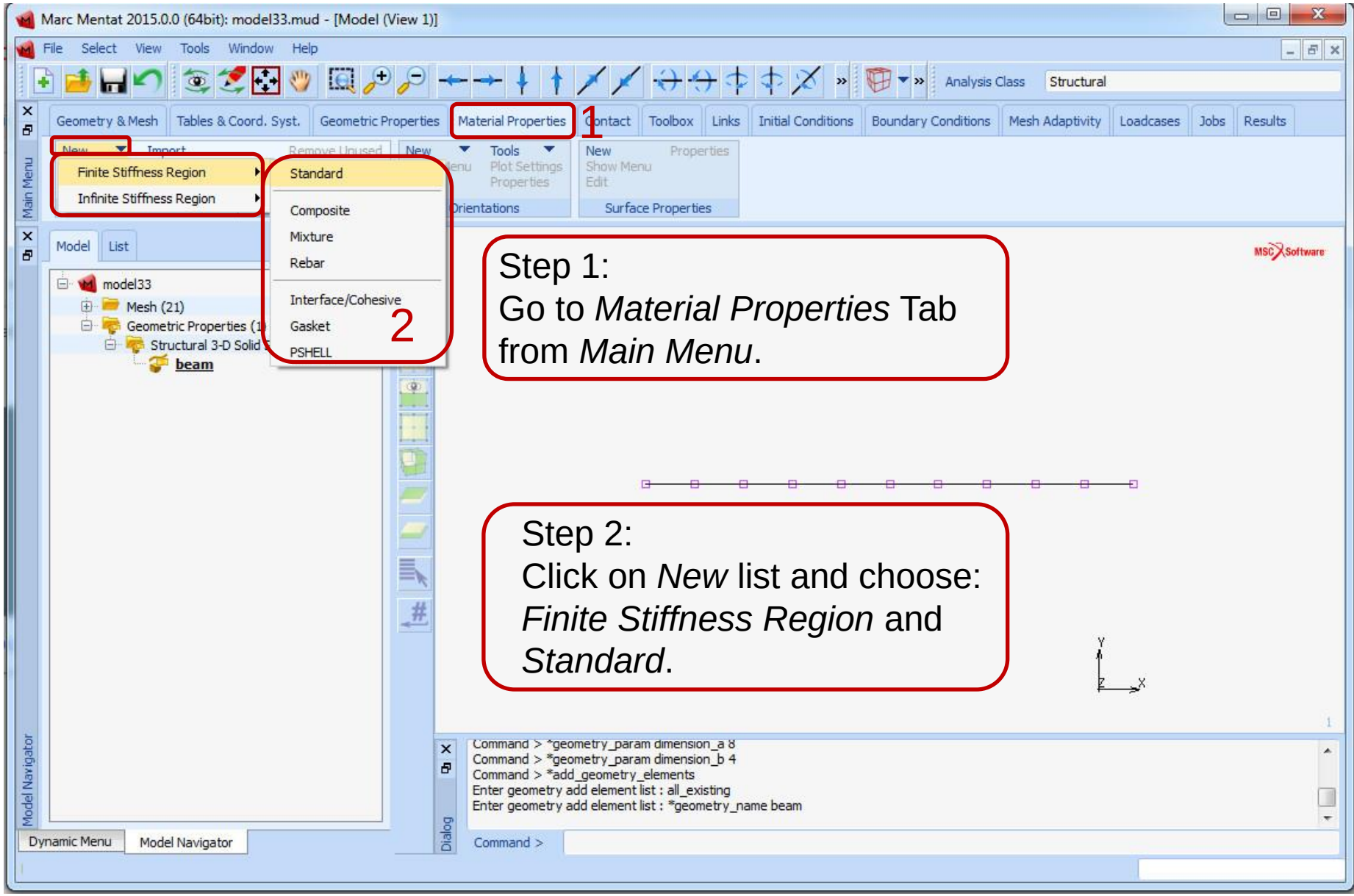
Step 2:
Input dimensions of scross section.

Step 3:
Click Add to prescribe geometric properties to elements.

Step 4:
Click all_existing – one should observe that properties have been added to 10 FEs 5 .



1D FEM solution



1D FEM solution

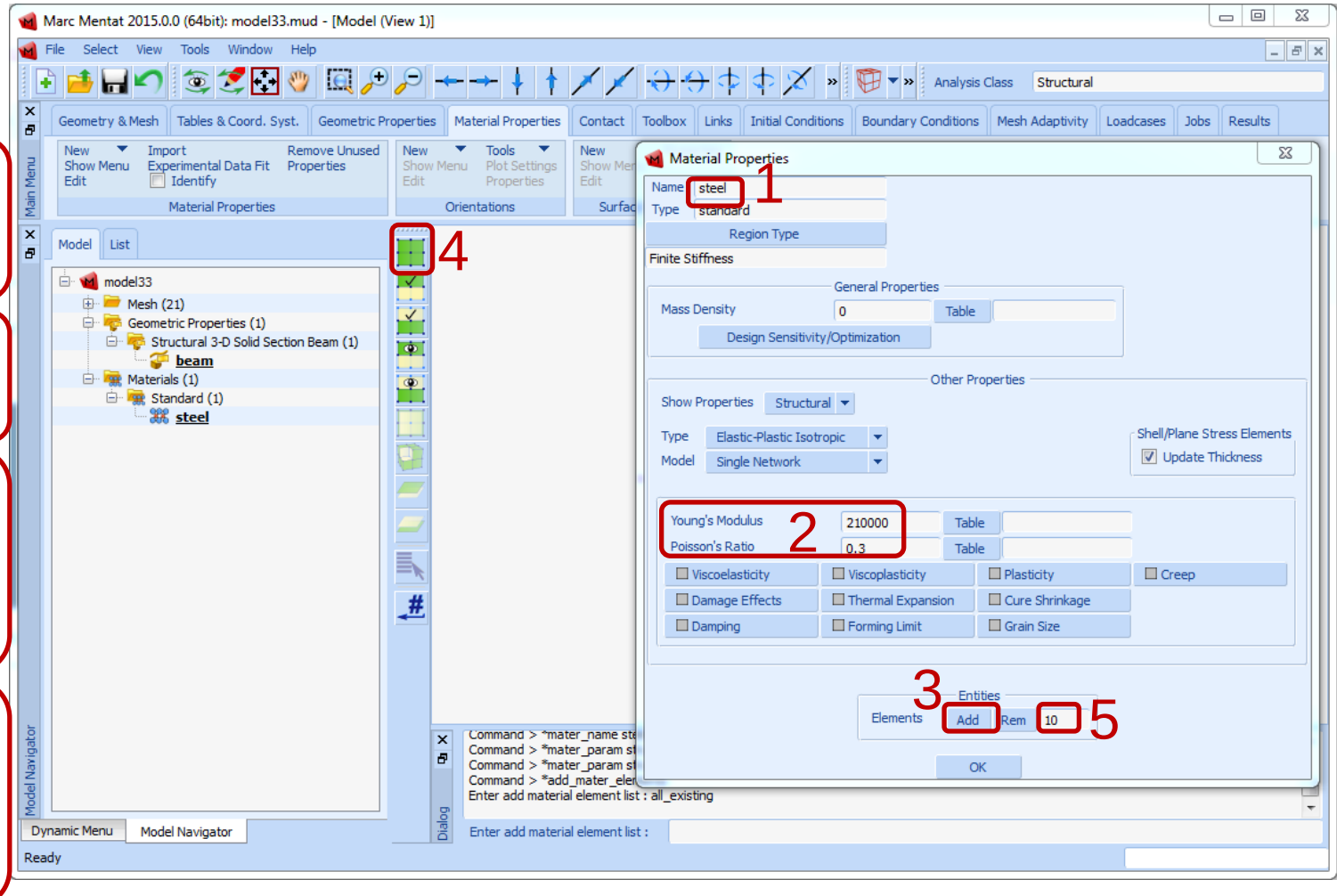
Action from previous slide opens Material Properties Menu.

Step 1:
Change *Name* to steel.

Step 2:
Input Young's Modulus equal 2.1e5 and Poisson's Ratio equal 0.3.

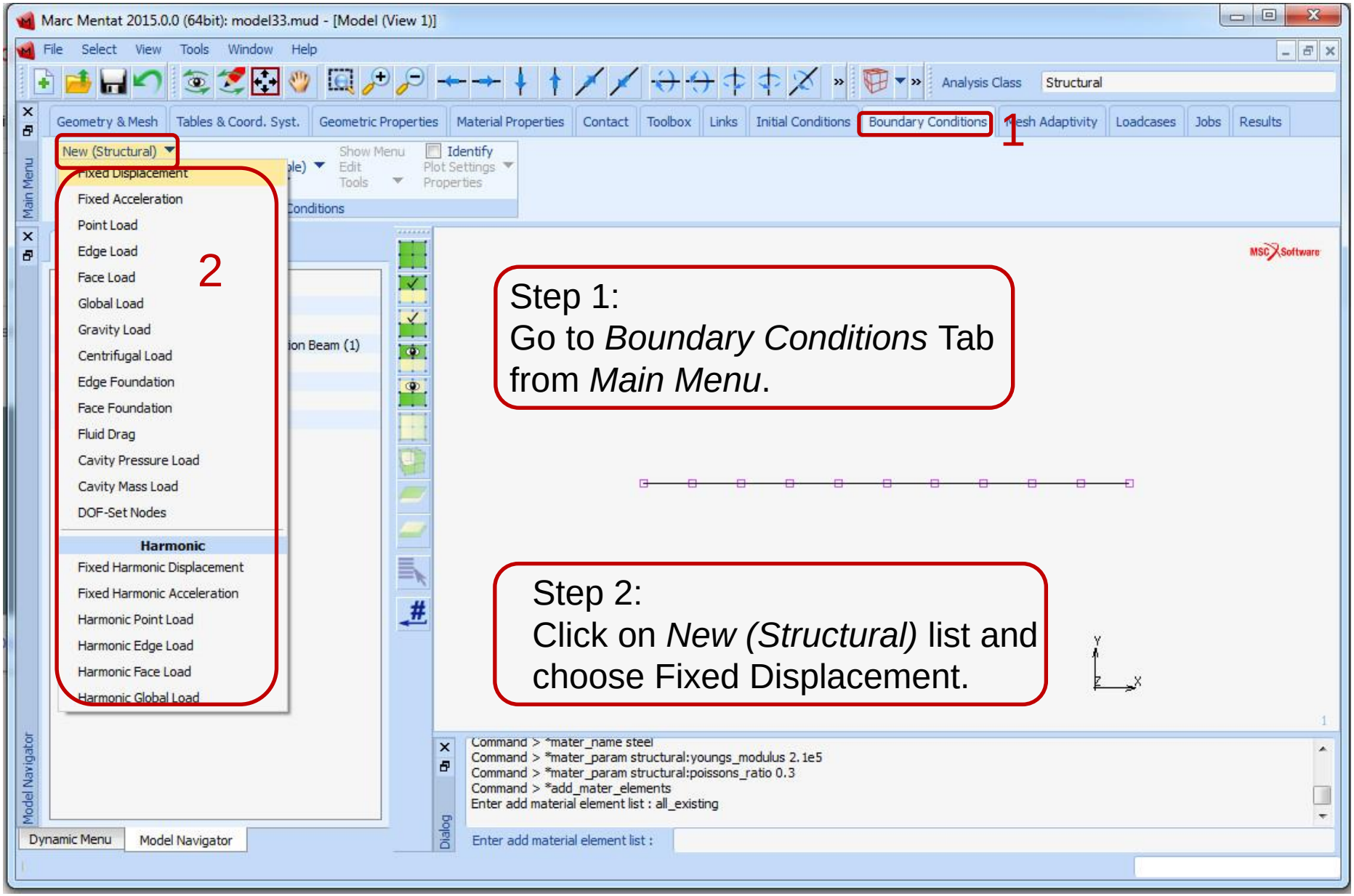
Step 3:
Click *Add* Button to prescribe material properties to elements.

Step 4:
Click all_existing – one should observe that properties have been added to 10 FEs 5 .



www.agh.edu.pl

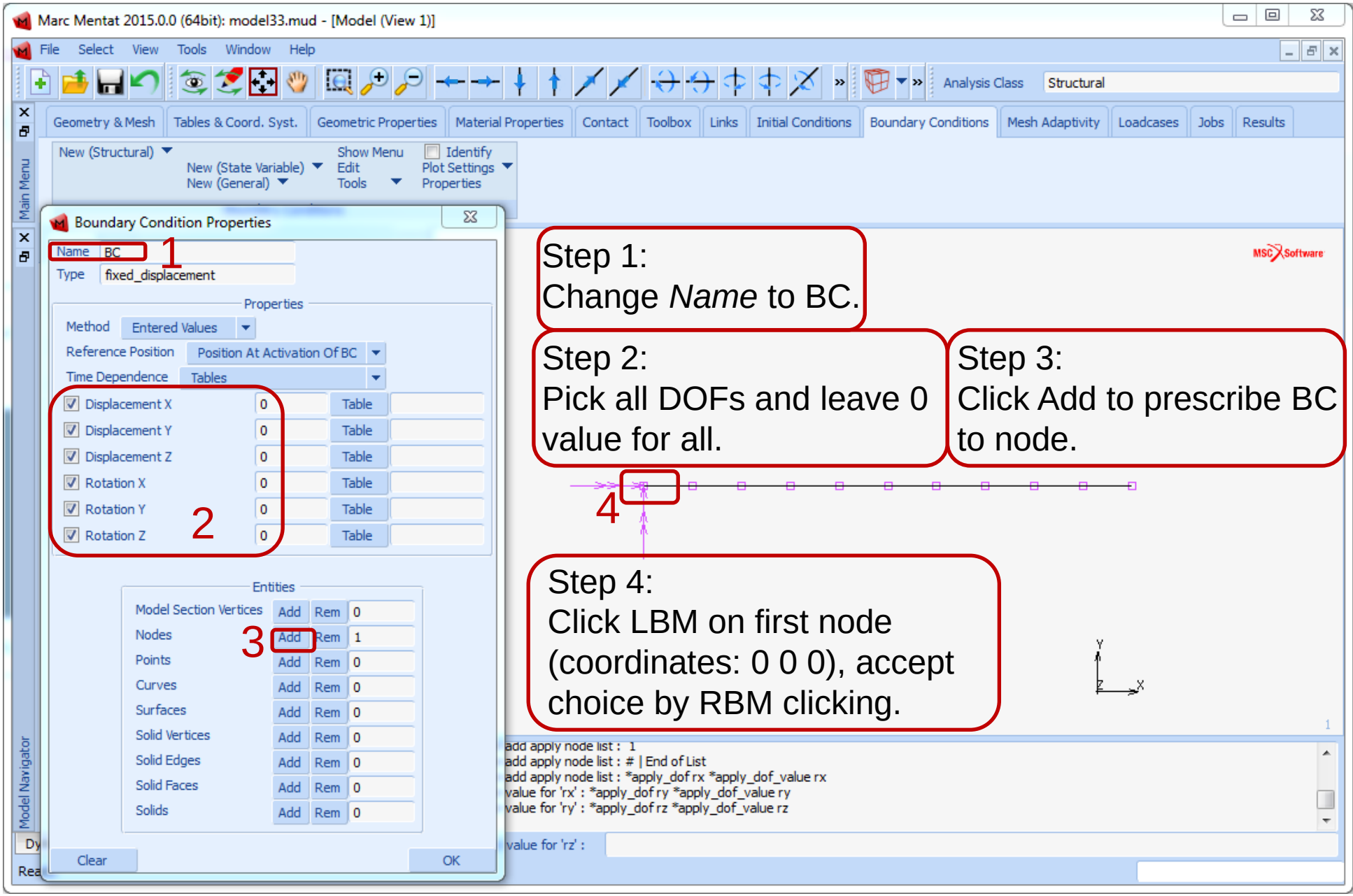
1D FEM solution



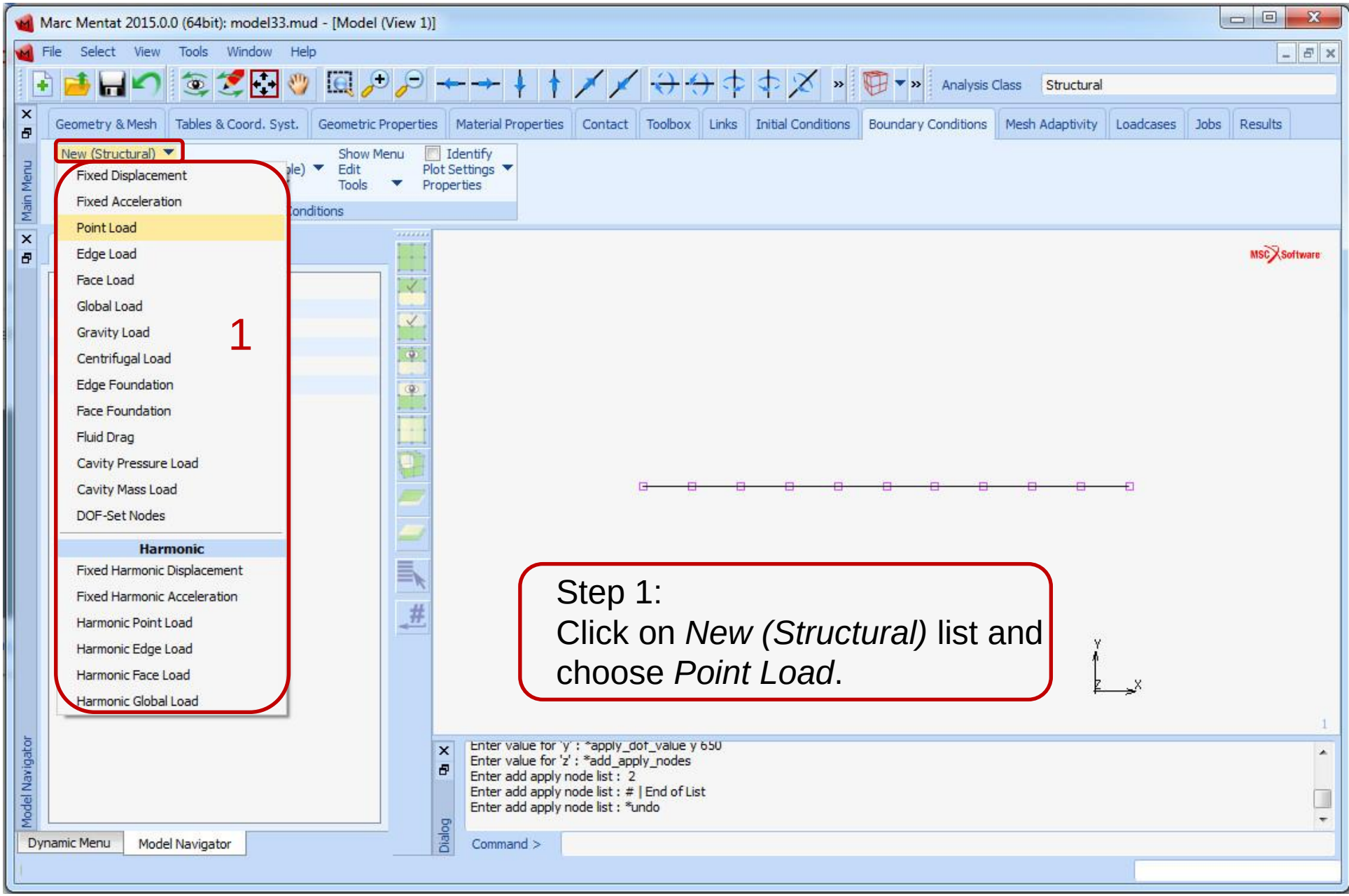
www.agh.edu.pl

1D FEM solution

Action from previous slide opens Boundary Condition Properties Menu.



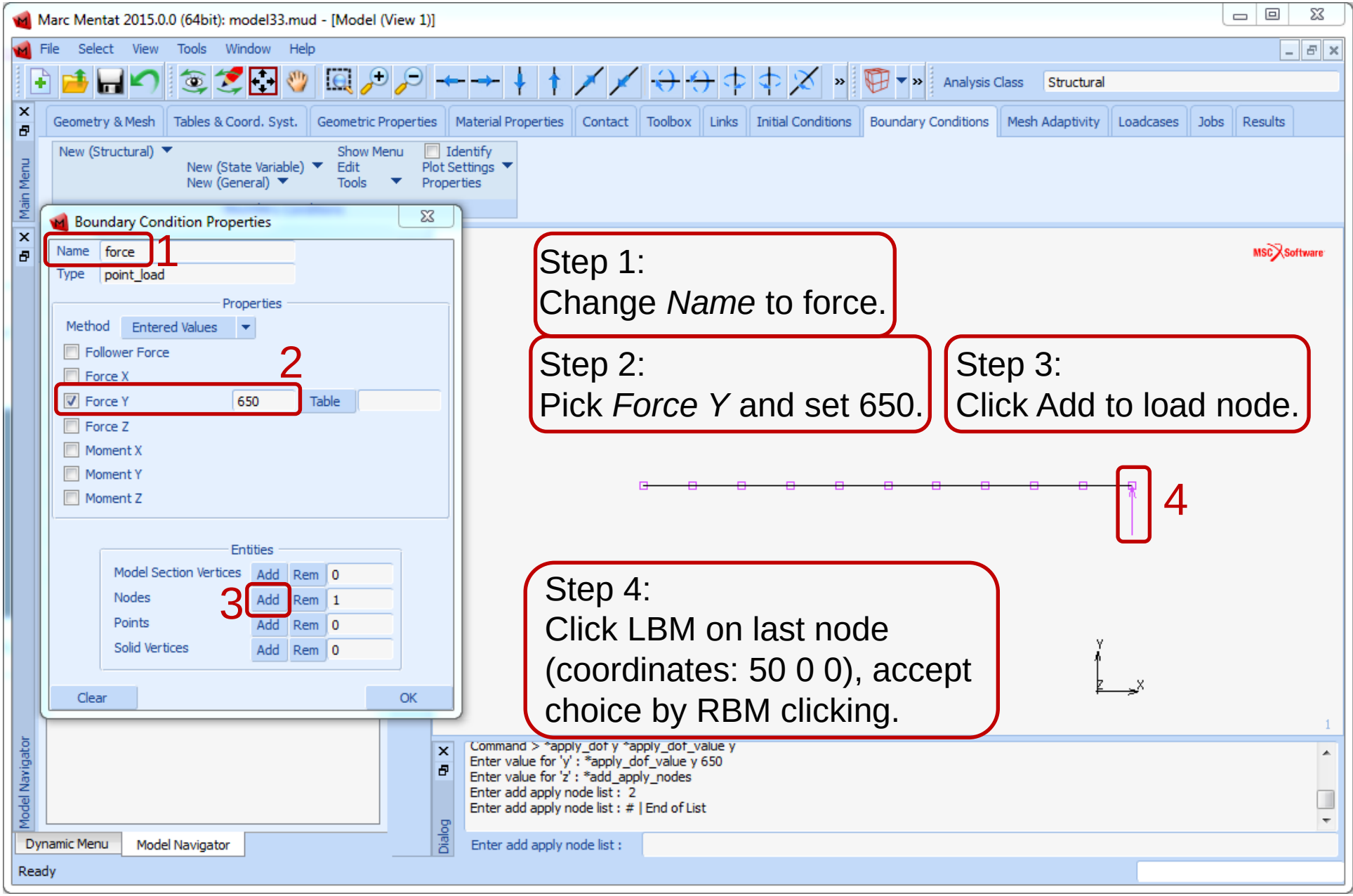
1D FEM solution



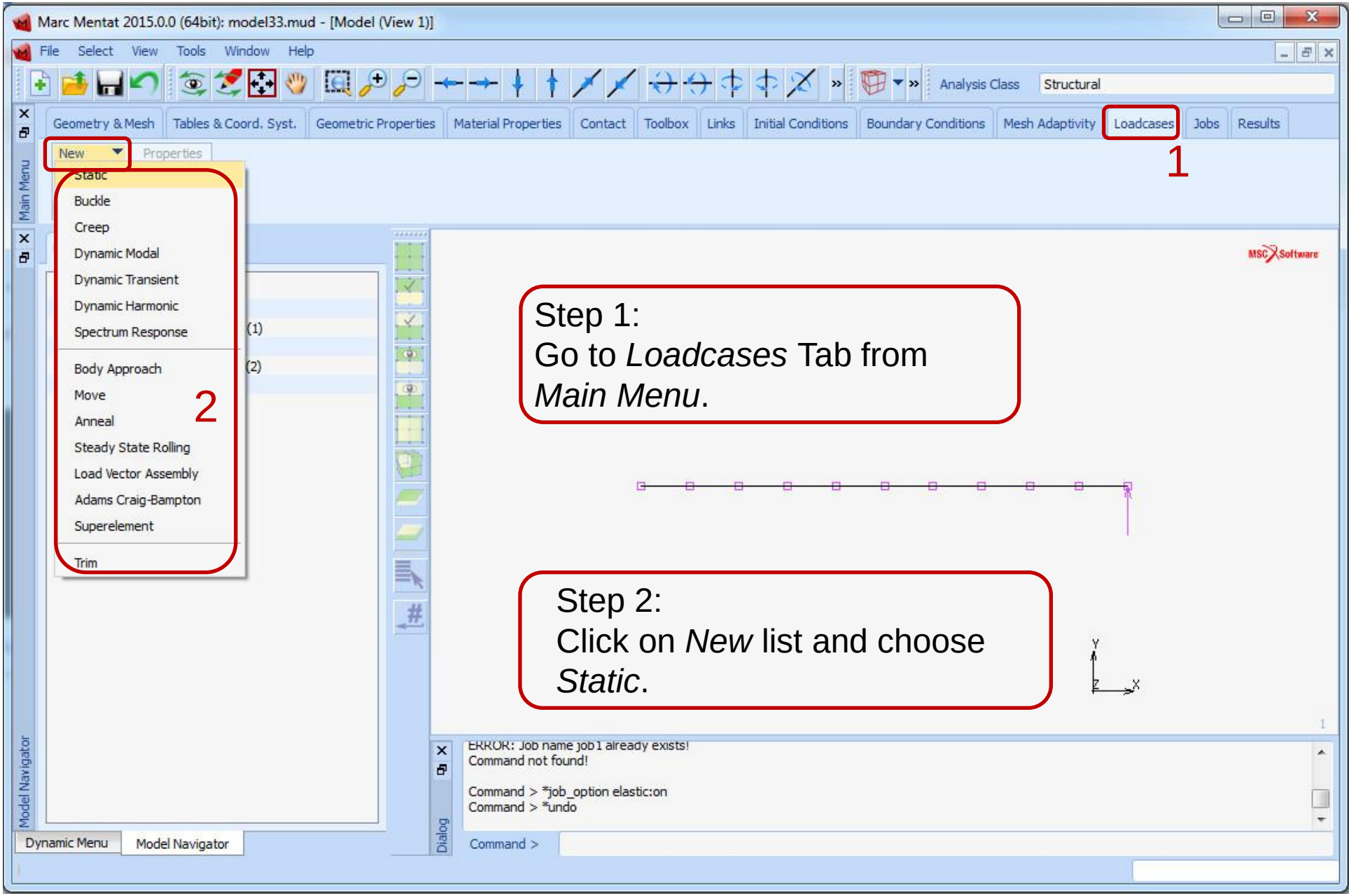
www.agh.edu.pl

1D FEM solution

Action from previous slide opens Boundary Condition Properties Menu.



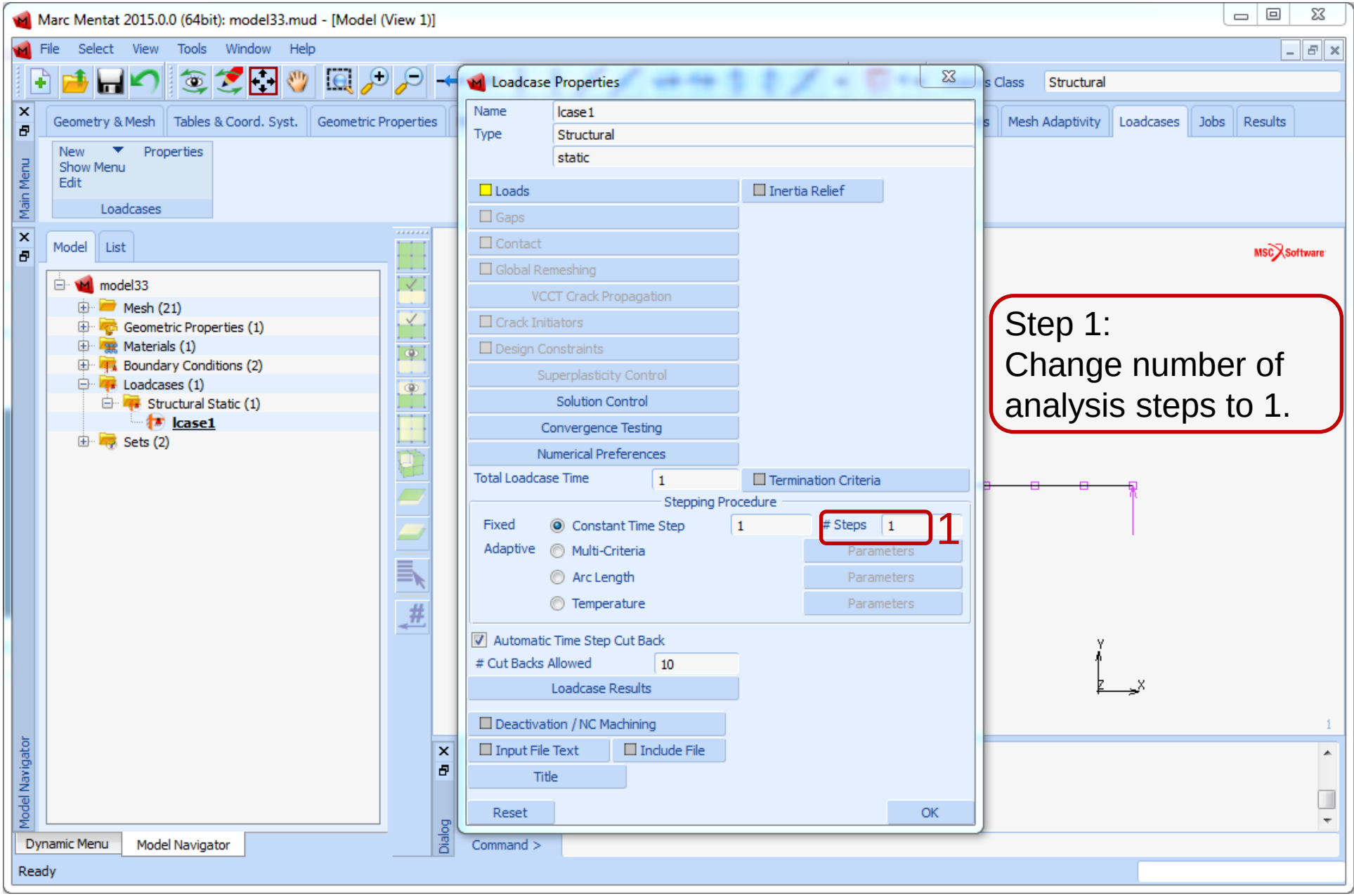
1D FEM solution



www.agh.edu.pl

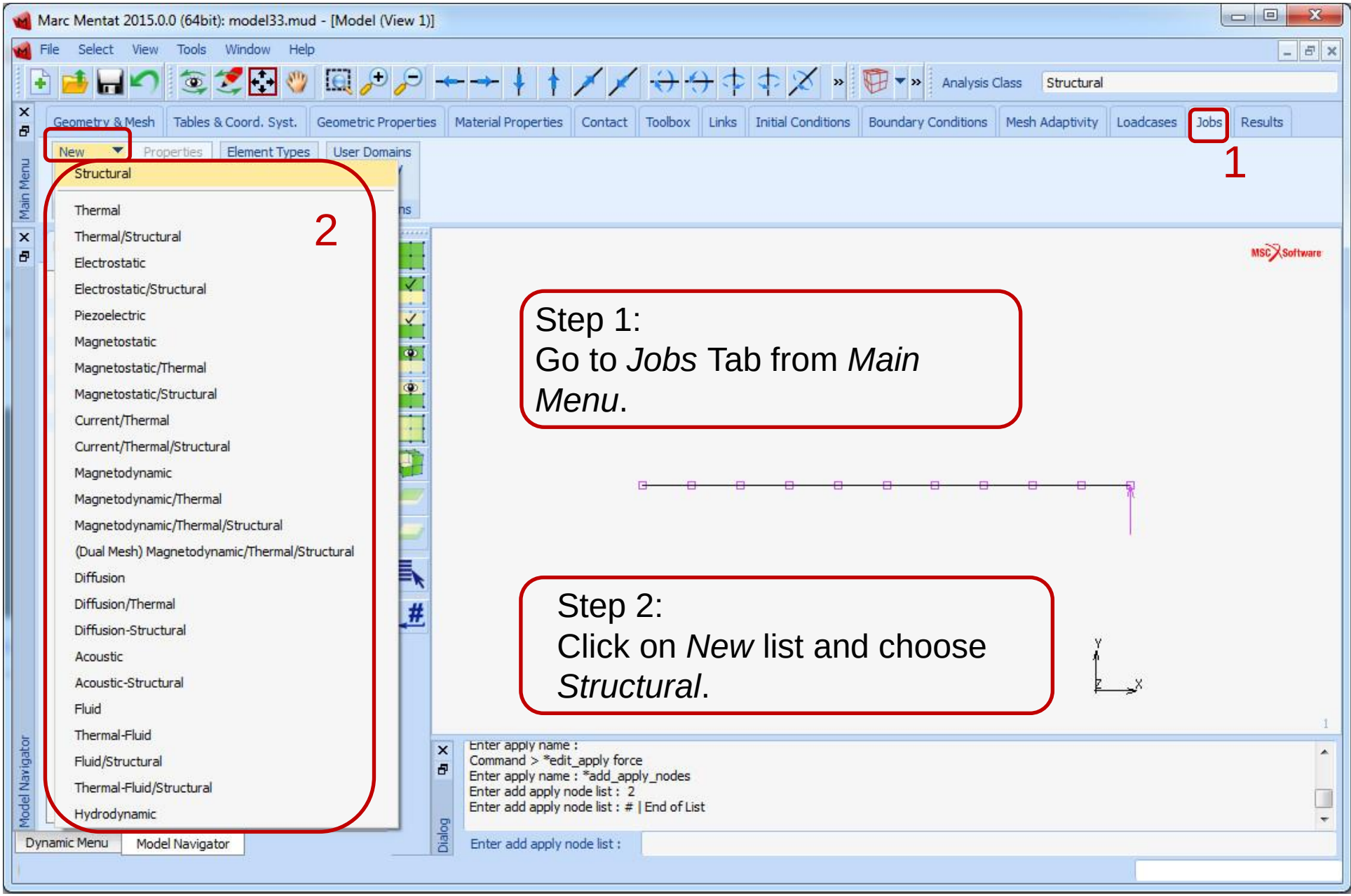
1D FEM solution

Action from previous slide opens *Loadcase Properties* Menu.



www.agh.edu.pl

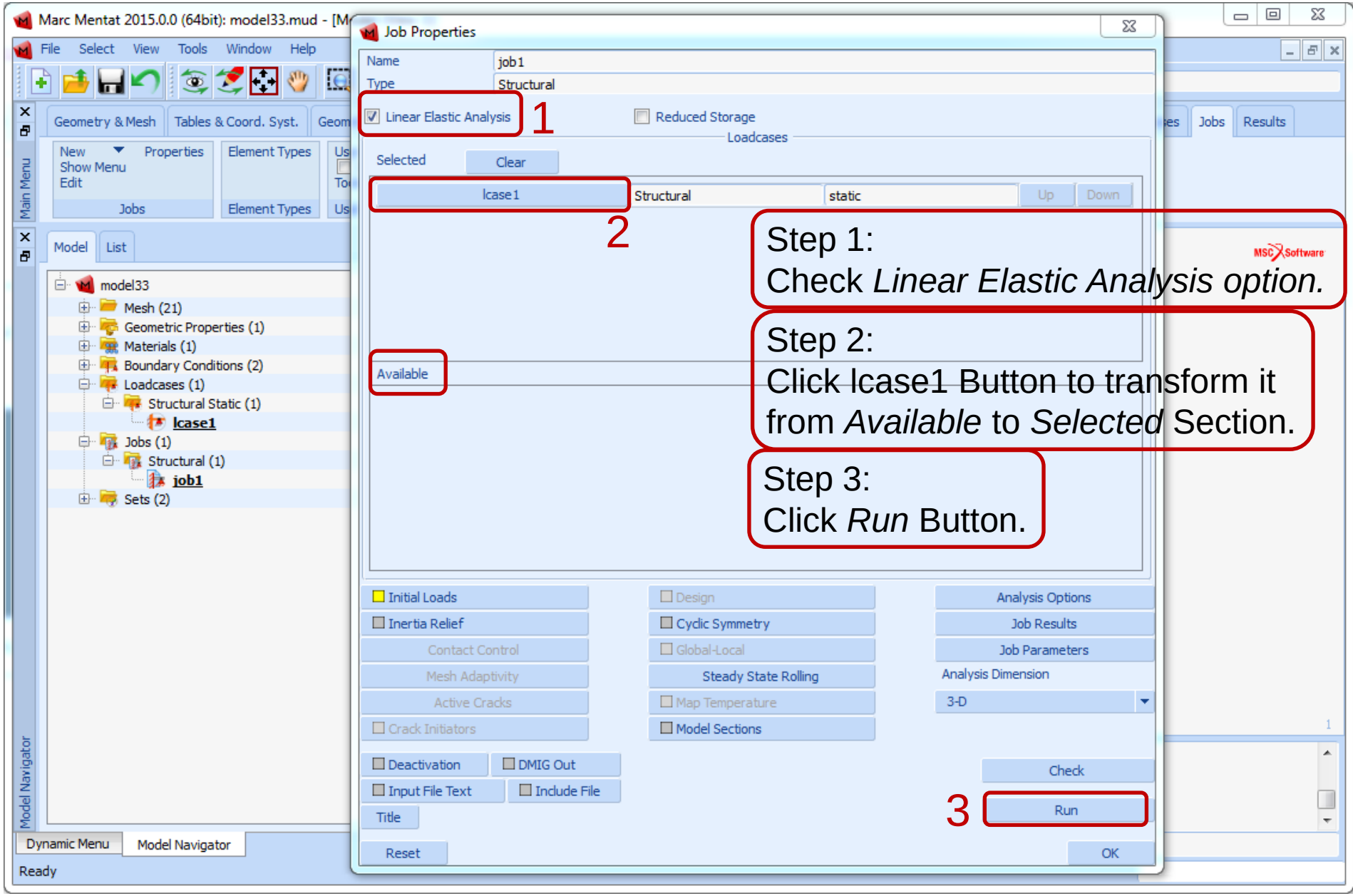
1D FEM solution



www.agh.edu.pl

1D FEM solution

Action from previous slide opens *Job Properties* Menu.



Step 1:
Check *Linear Elastic Analysis option.*

Step 2:
Click lcase1 Button to transform it from *Available* to *Selected* Section.

Step 3:
Click *Run* Button.

1D FEM solution

Action from previous slide opens *Run Job* Menu.

Step 1:
Save your model by clicking on the Button.

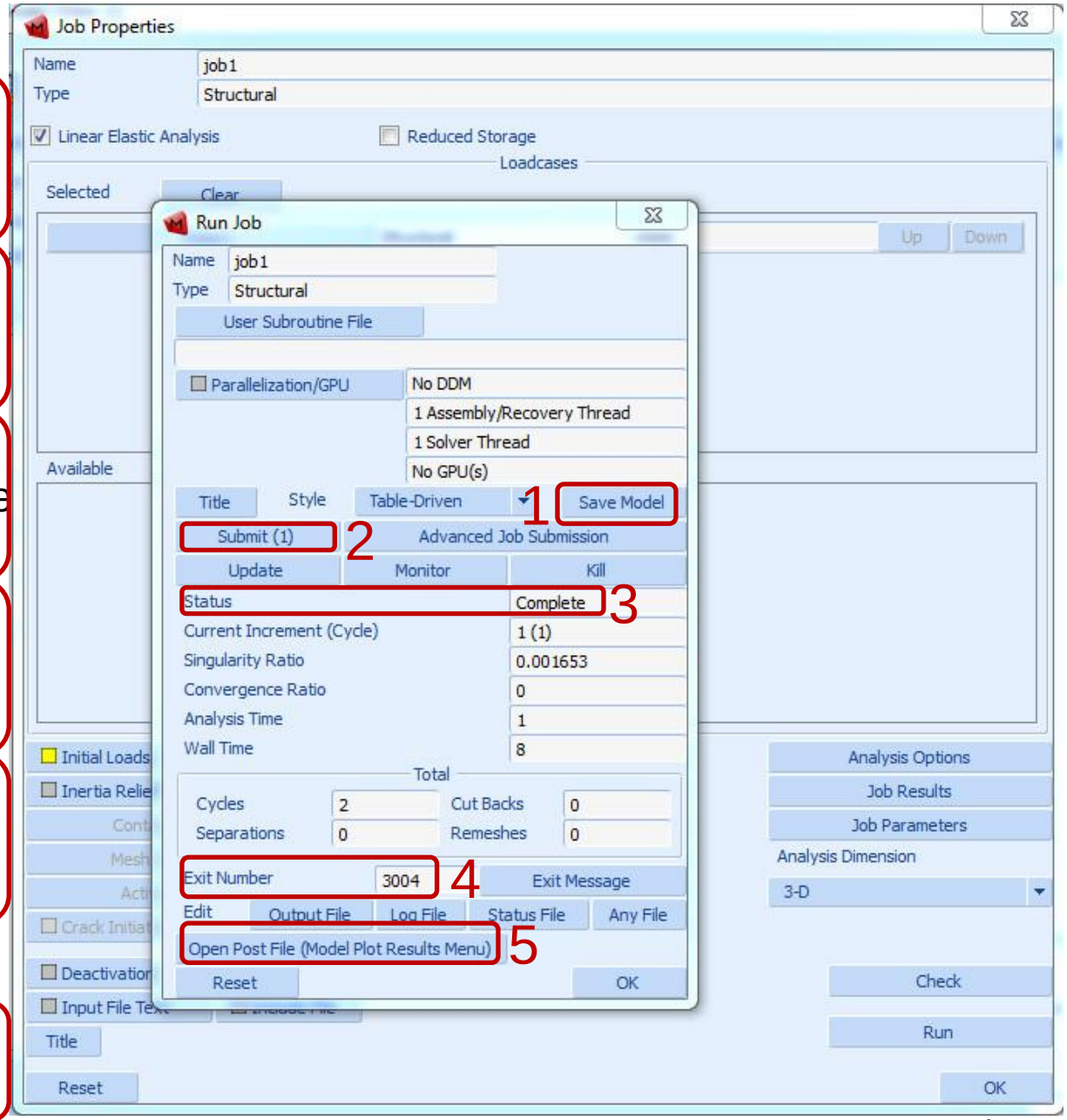
Step 2:
Click *Submit(1)* Button to run simulation.

Step 3:
The simulation was finished when the status change to *Complete*.

Step 4:
The simulation return results when *Exit Number* is 3004.

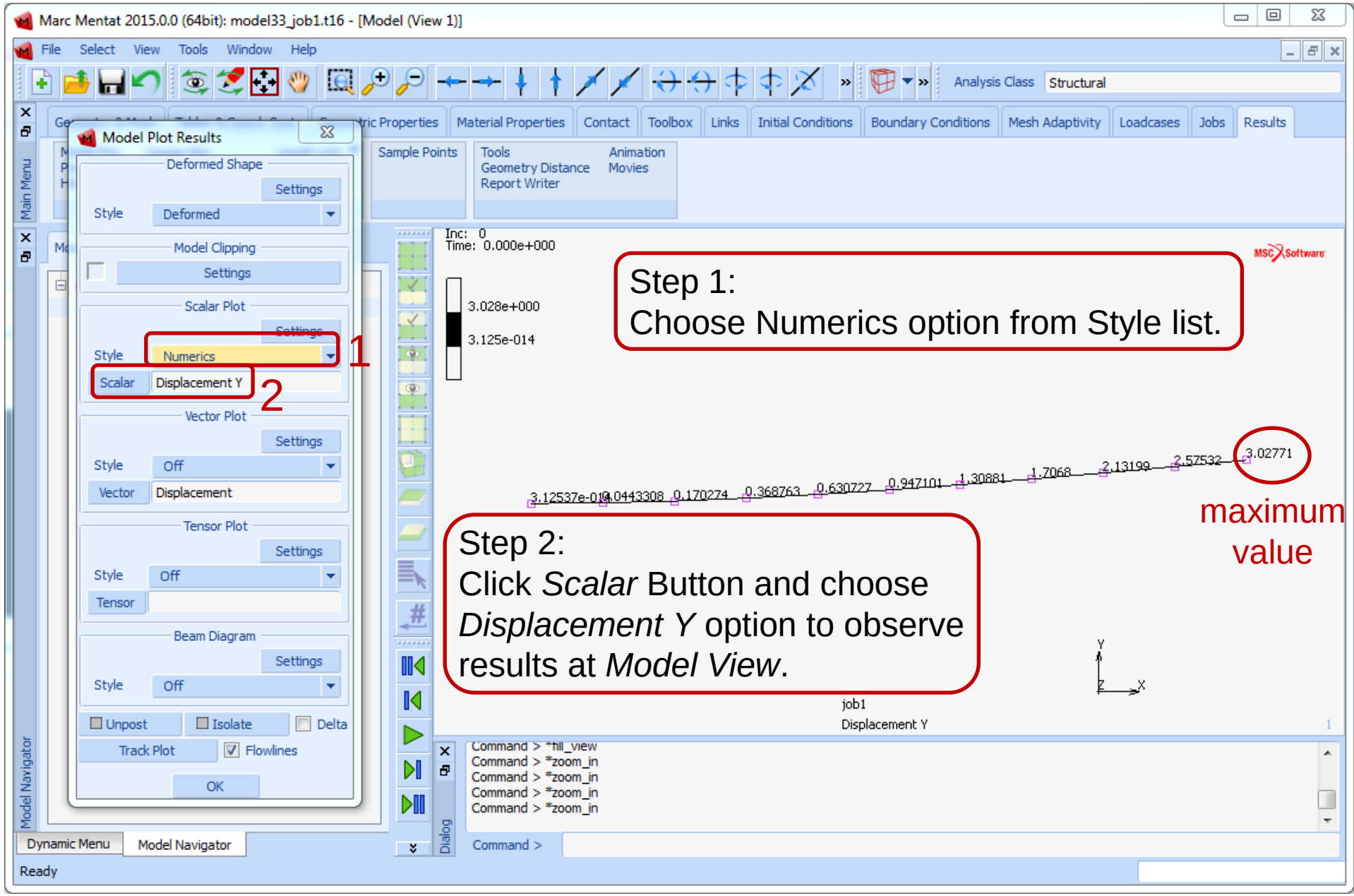
Step 5:
Click *Open Post File (Model Plot Results Menu)* Button to plot results.

Here is last chance to save model until one does not do the step 5.



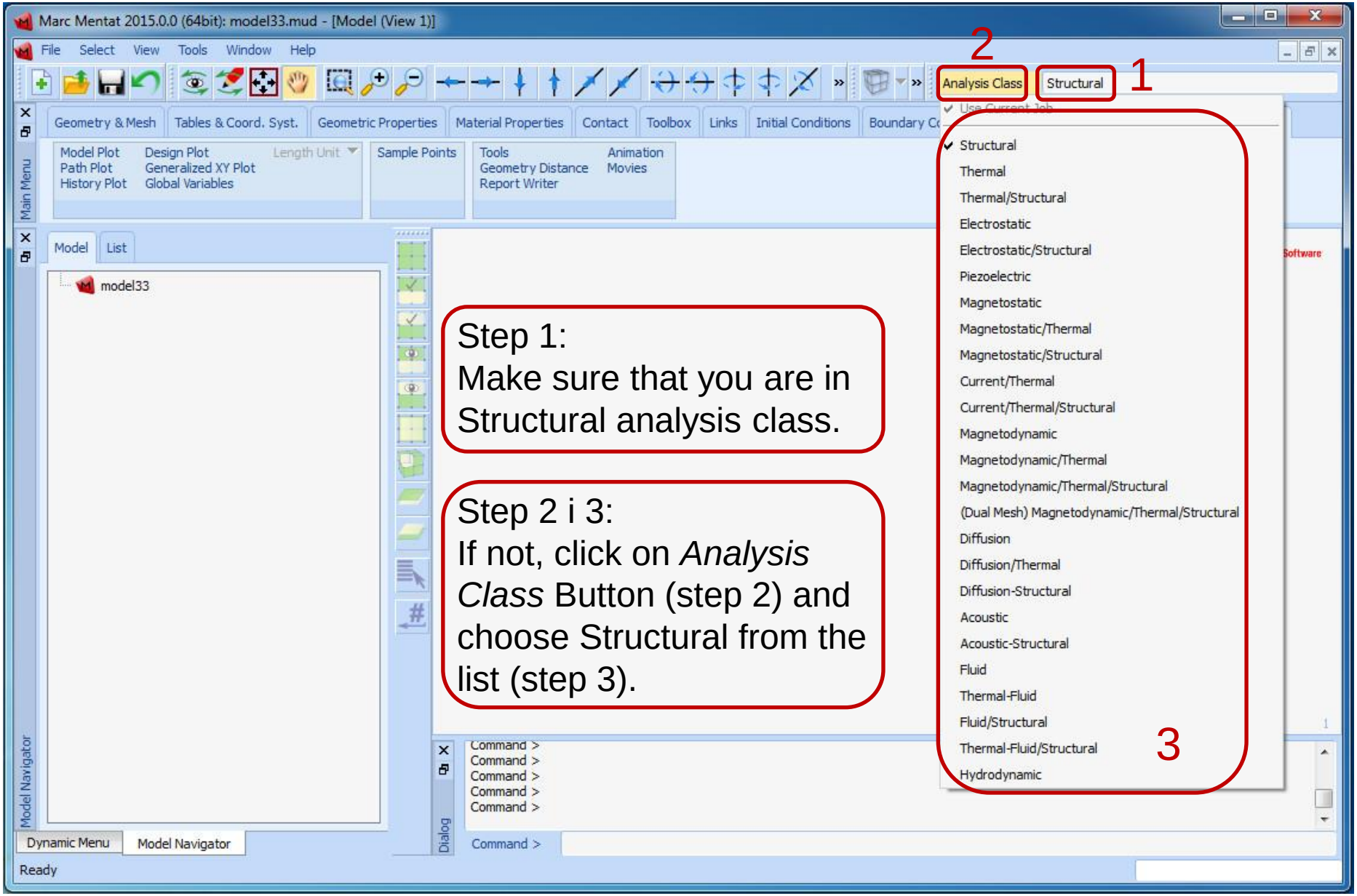
1D FEM solution

Action from previous slide opens *Model Plot Results* Menu.



www.agh.edu.pl

3D FEM solution



3D FEM solution

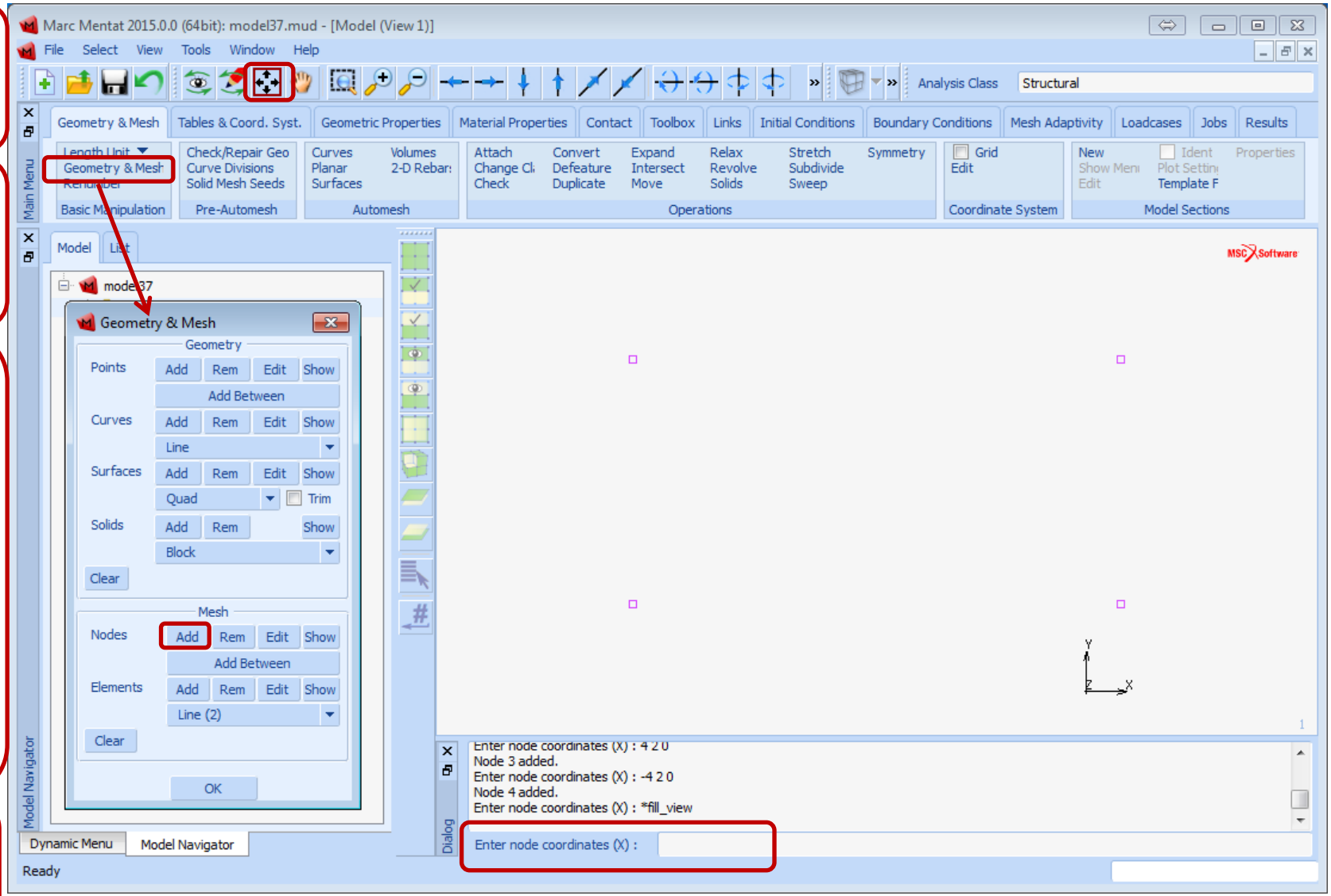
Create model in one of the known you method. In this case we use the first method described in *Geometry & Mesh* tutorial.

Step 1:
Open *Geometry & Mesh* Menu.

Step 2:
Click *Add* Button to add 4 nodes.

Step 3:
Input coordinates data to *Command Window*:
-> -4 -2 0 (Enter)
-> 4 -2 0 (Enter)
-> 4 2 0 (Enter)
-> -4 2 0 (Enter)

Step 4:
Click *Fill view* Button to show created nodes.



3D FEM solution

Create model in one of the known you method. In this case we use the first method described in *Geometry & Mesh* tutorial.

Step 1:
Choose *Quad(4)* option form list.

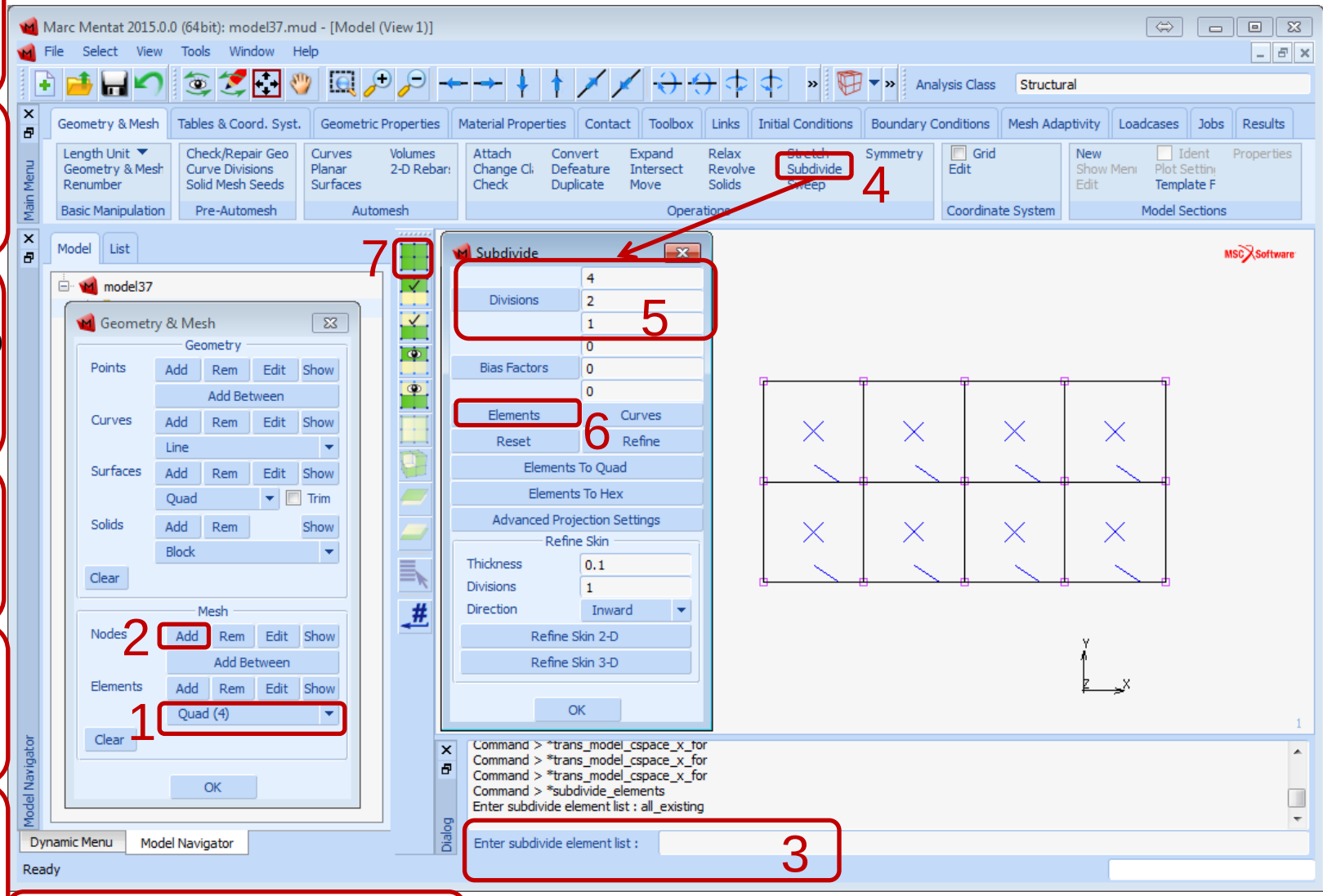
Step 2:
Click *Add* Button to add FE.

Step 3:
Input node indexes to *Command Window*:
-> 1 2 3 4 (Enter)

Step 4:
Open *Subdivide Menu*.

Step 5:
Set *Divisions* parameters.

Step 6:
Click *Elements* Button to divide FE.



Step 7:
Click *all_existing* Button.

www.agh.edu.pl

3D FEM solution

Step 1:
Open *Expand* Menu.

Step 2:
Set expand options.

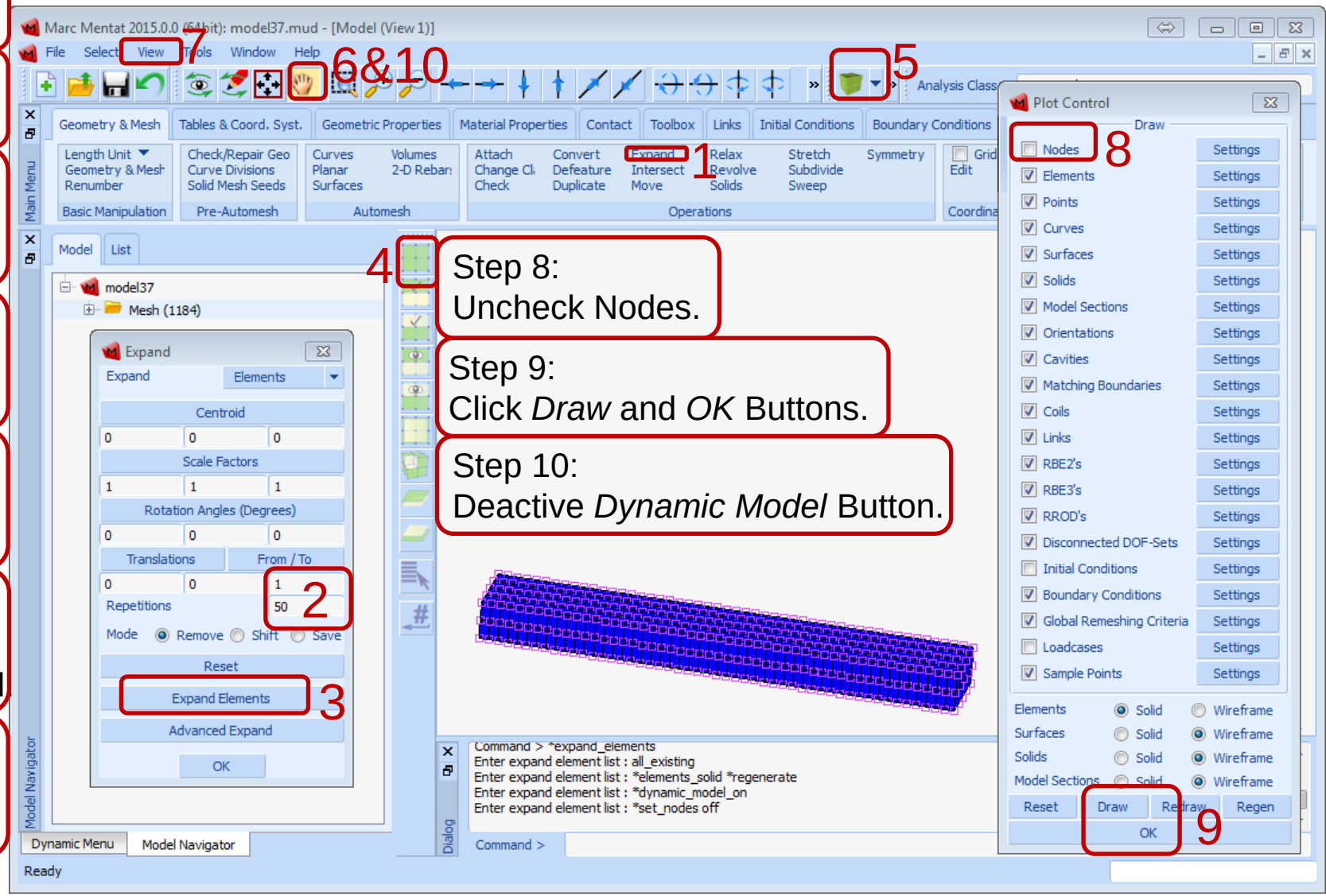
Step 3:
Click *Expand Elements* Button.

Step 4:
Click *all_existing* Button.

Step 5: Change the model presentation to *Elements Solid*.

Step 6:
Active *Dynamic Model* Button and rotate model

Step 7:
Click *View* and *Plot Control*….

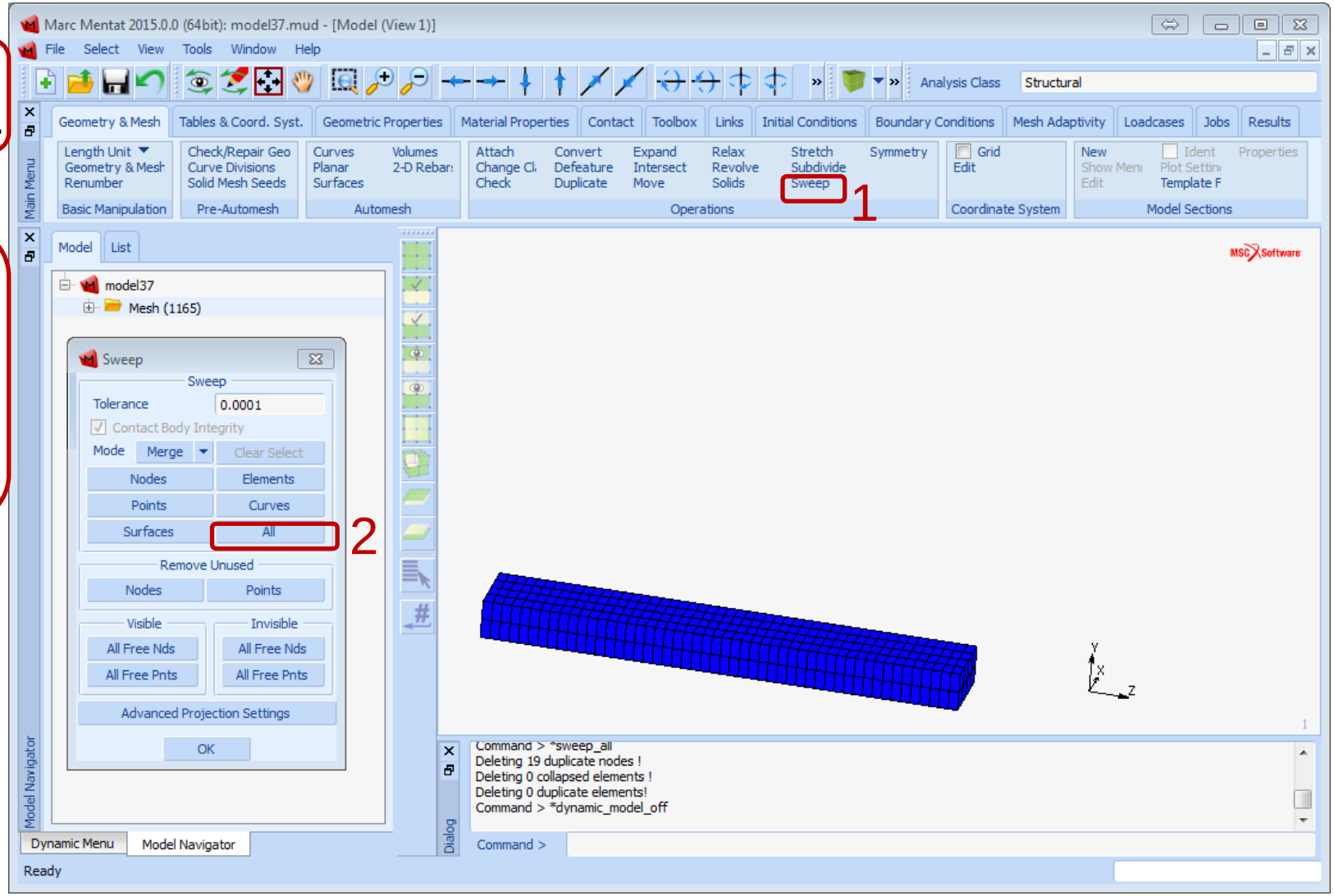


3D FEM solution

Step 1:
Open *Sweep* Menu.

Step 2:
Click *All* Button.

Note: 19 nodes have been deleted.

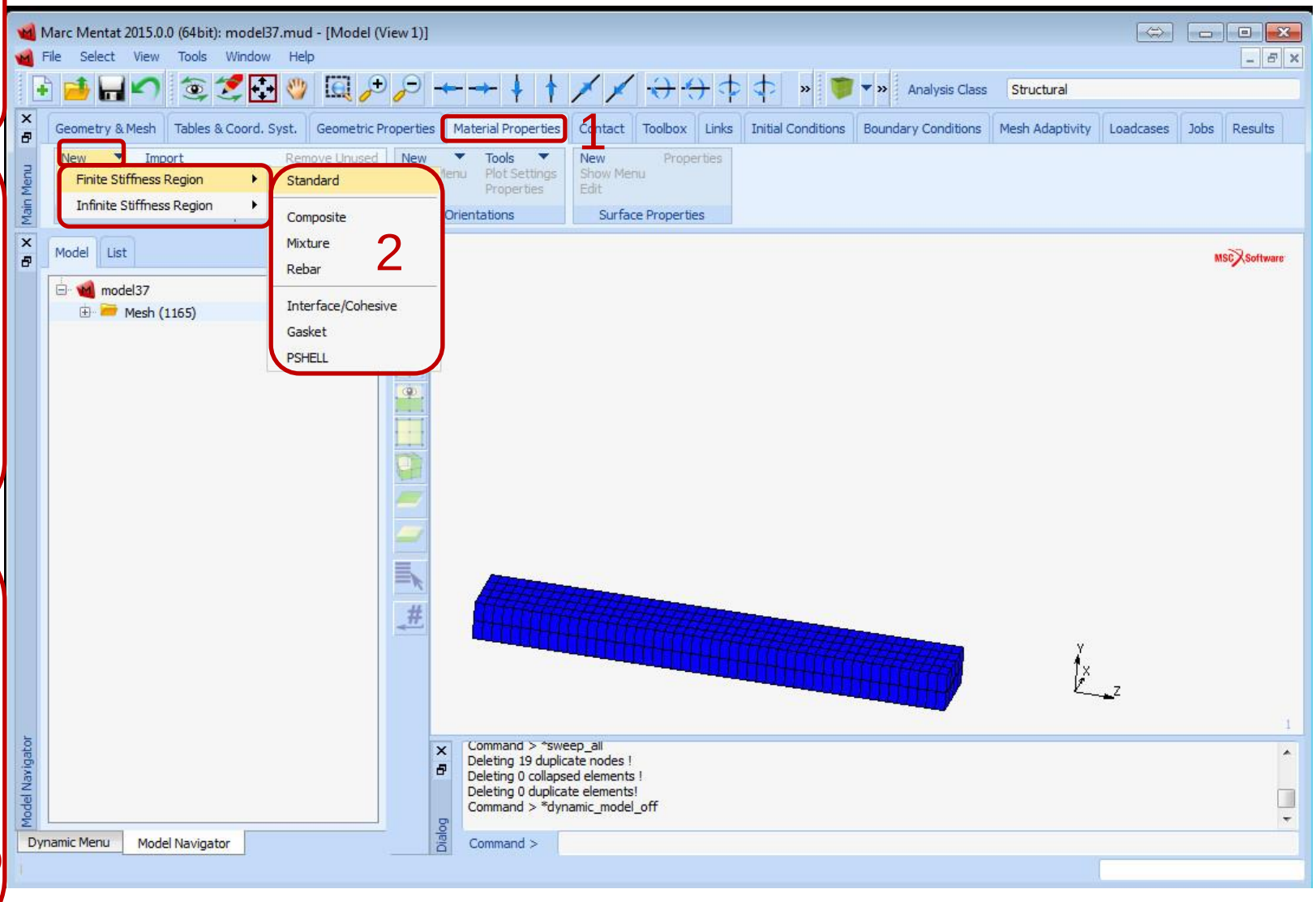


3D FEMsolution

Step 1:
Go to *Material Properties* Tab from *Main Menu*.

Step 2:
Click on New (Structural) list and choose: Finite Stiffness Region and next Standard option.

Note:
During 3D modeling one use different material model than during 2D modeling. Marc shows material models which match to the used FE type.



www.agh.edu.pl

3D FEM solution

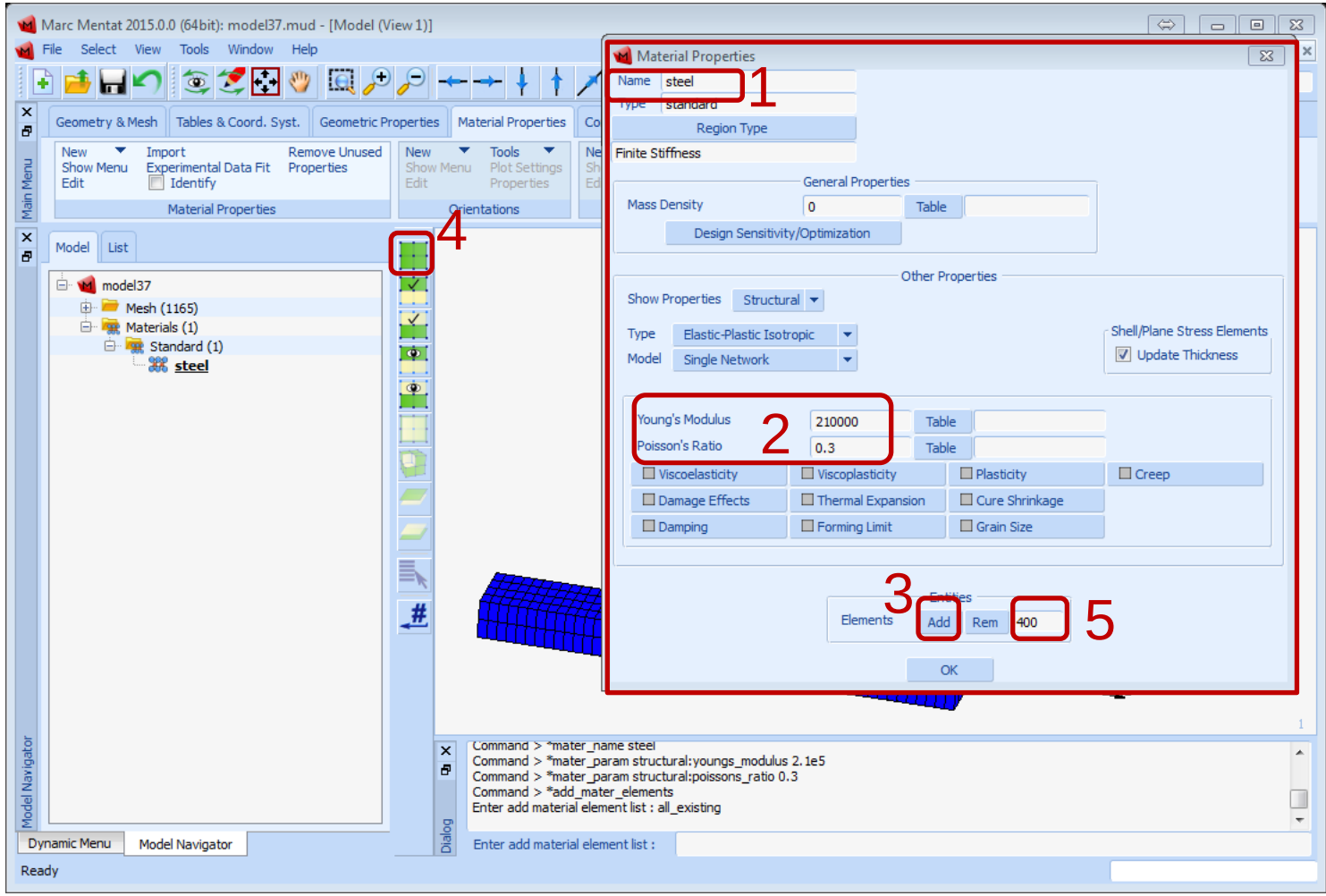
Action from previous slide open *Material Properties* Menu.

Step 1:
Change *Name* to steel.

Step 2:
Input Young's Modulus equal 2.1e5 and Poisson's Ratio equal 0.3.

Step 3:
Click Add to prescribe material properties to elements.

Step 4:
Click all_existing – one should observe that properties have been added to 400 FEs 5 .



www.agh.edu.pl

3D FEM solution

Step 1:
Go to *Boundary Conditions* Tab from Main Menu.

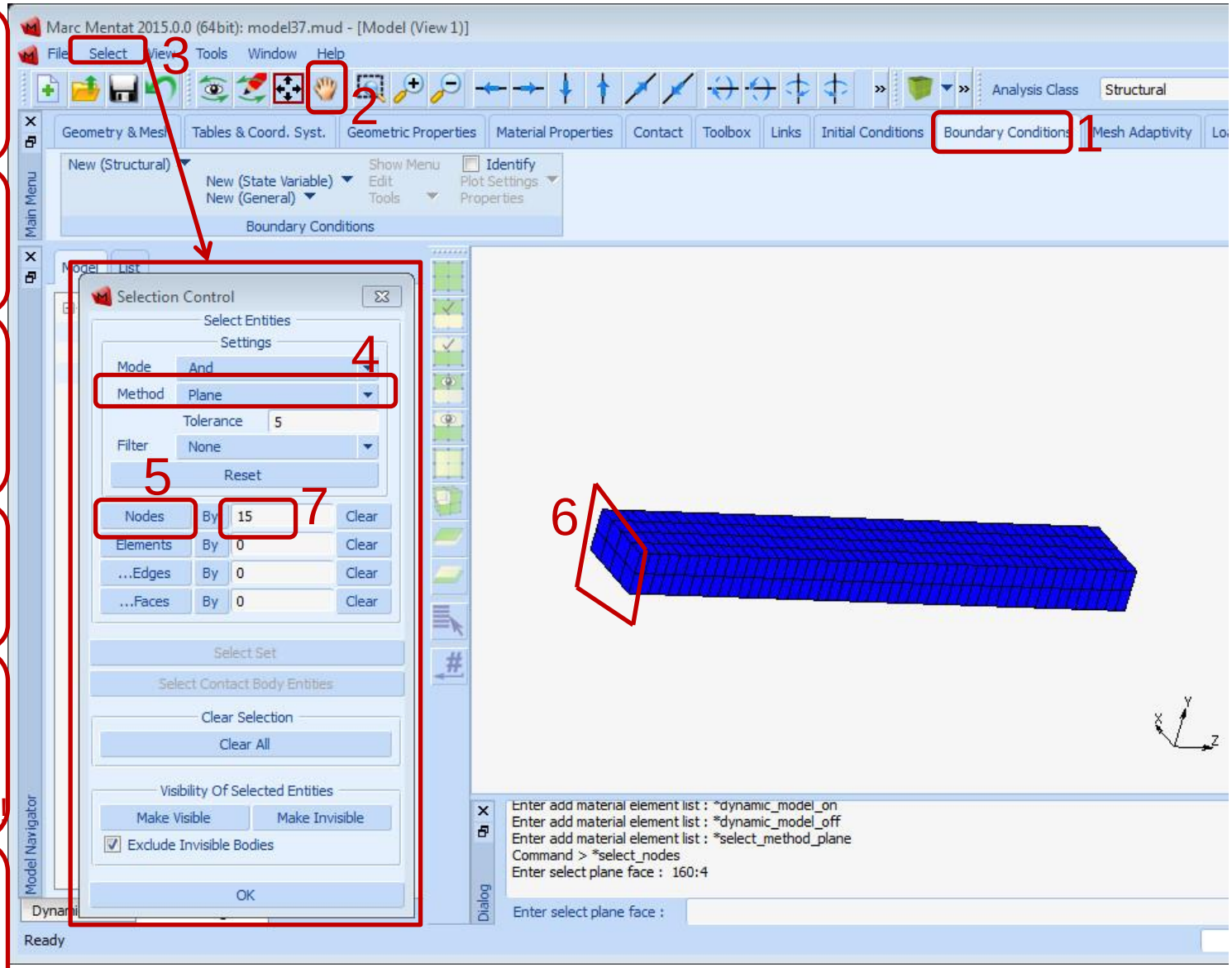
Step 2:
Active *Dynamic Model* Button and rotate model.

Step 3:
From *Select* list choose *Selection Control…* option – new menu has opened.

Step 4:
Change *Method* to *Plane* option.

Step 5:
Click *Nodes* Button to choose nodes which will be fixed – deactivate *Dynamic Model* Button!

Step 6:
Pick one of nodes which lie on XY plane – you should note 15 selected nodes 7.



3D FEM solution

Step 1:
Click on *New (Structural)* list and choose *Fixed Displacement* option.

Step 2:
Change name to *fixed_nodes*.

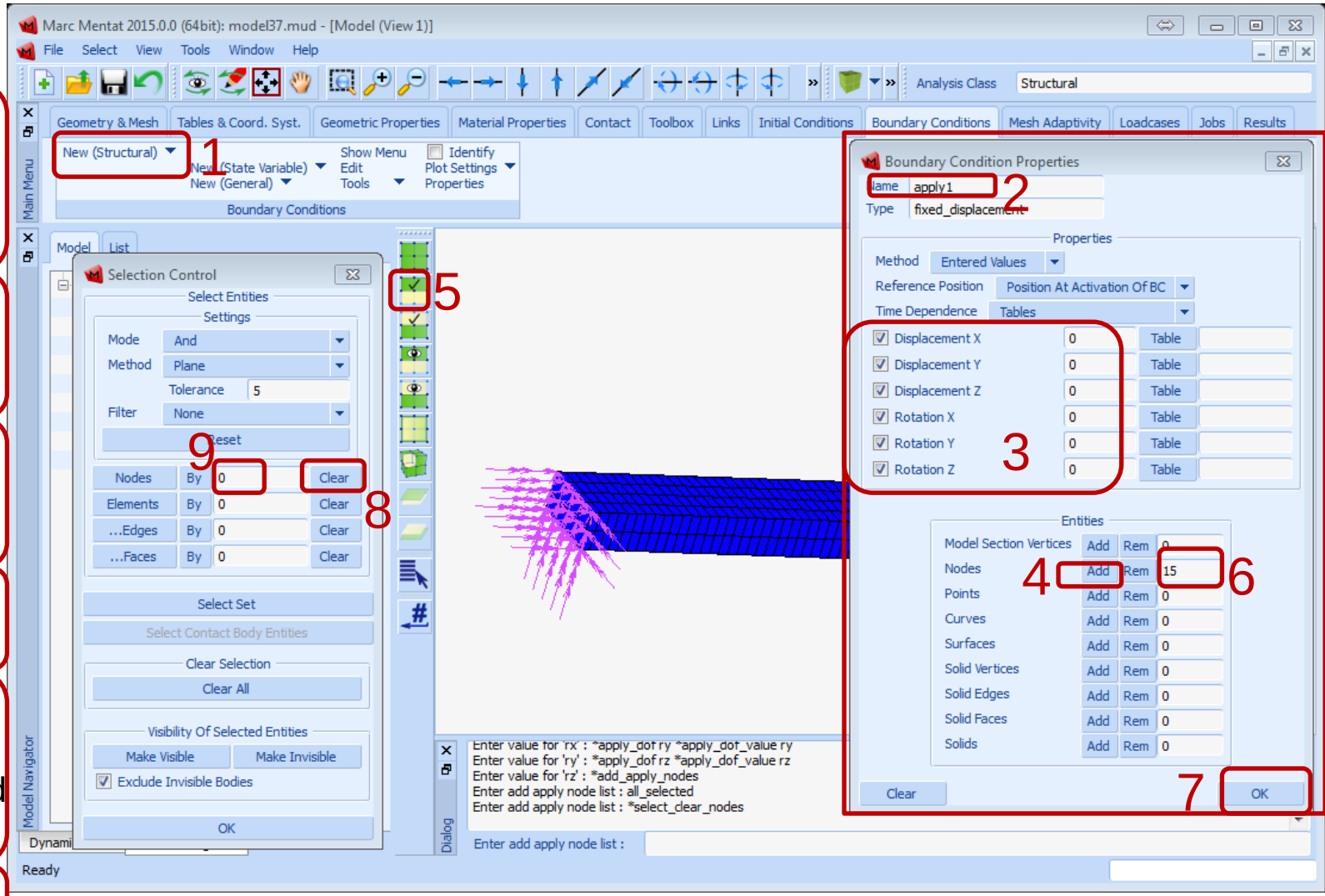
Step 3:
Tick all DOFs and leave value 0.

Step 4:
Click *Add* Button.

Step 5:
Click *all_selected* Button – 15 nodes should be selected 6.

Step 7:
Click OK Button.

Step 8:
Click Clear Button – now none nodes are selected 9.



www.agh.edu.pl

3D FEM solution

Step 1:
Click on *New (Structural)* list and choose *Point Load* option.

Step 2:
Change name to *force*.

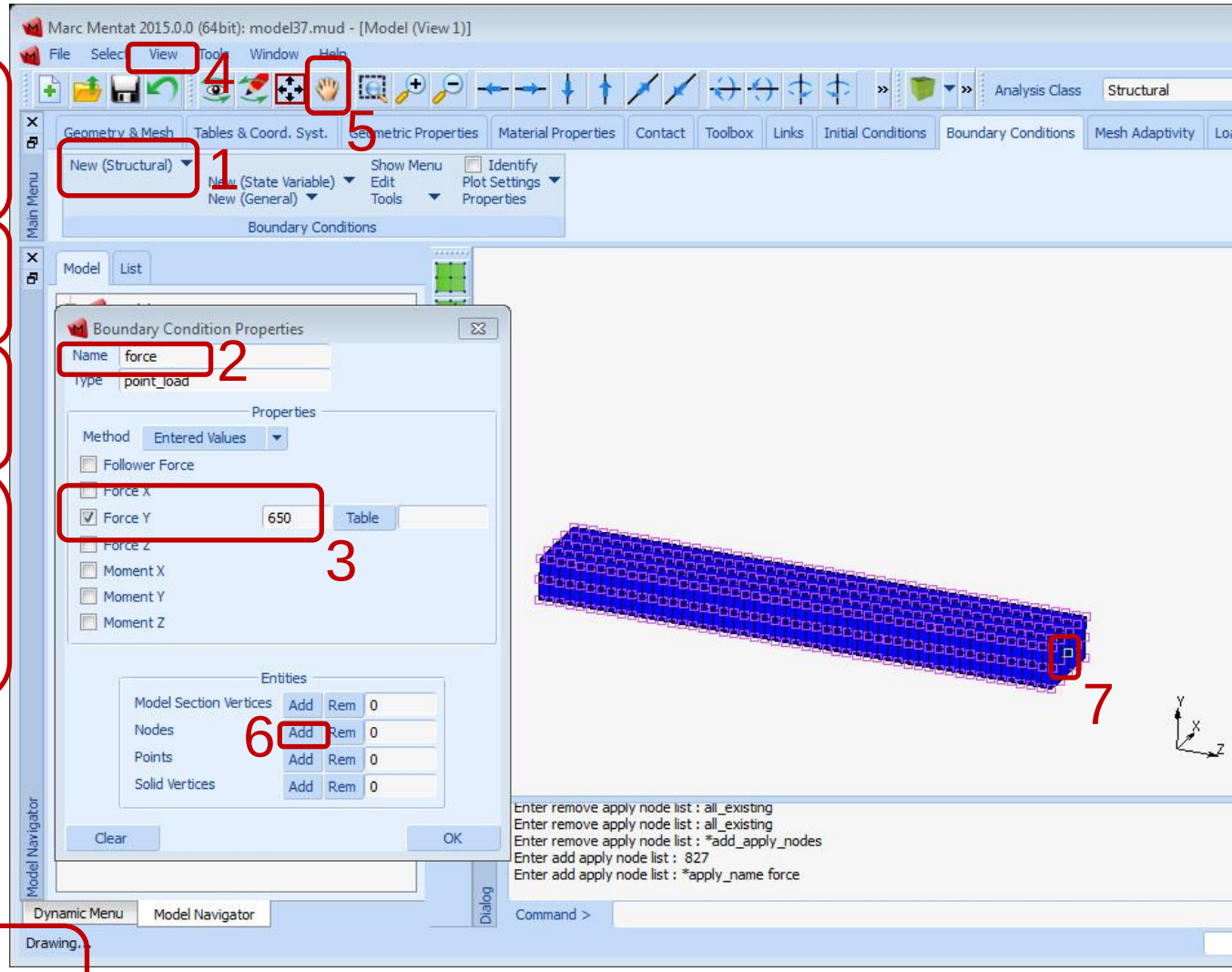
Step 3:
Tick *Force Y* and input *650*.

Step 4:
Click *View* list,choose *Model Entity Types* and tick *Nodes* option.

Step 5:
Active *Dynamic Model* Button and rotate model as on screenshot.
Deactivate *Dynamic Model* Button.

Step 6:
Click *Add* Button to apply force to node.

Step 7:
Click LBM on the node (the selected node change color to green). Accept choice by RBM clicking.



3D FEM solution

REPEAT OPERATIONS FROM SLIDES 16-20

TO DEFINED

LOADCASES, JOBS, RUN SIMULATION

AND OPEN RESULTS.

www.agh.edu.pl

3D FEM solution

Step 1:
Choose *Contour Bands* option from *Style* list.

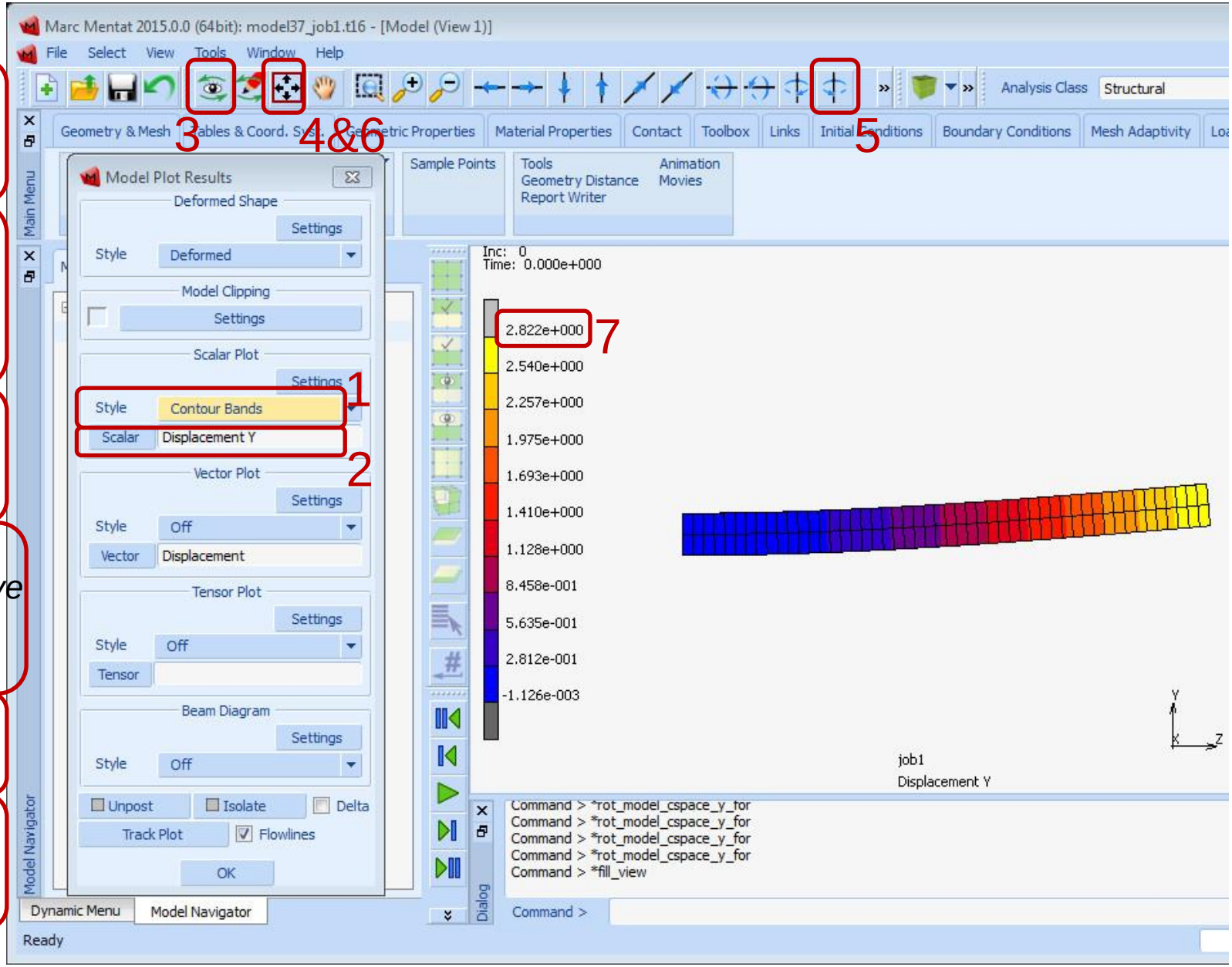
Step 2:
Click *Scalar* Button and choose *Displacement Y* option.

Step 3 & 4:
Click *Reset view* Button and next *Fill view* Button.

Step 5:
Click *Rotate about positive Y-axis* Button until you observe YZ plane.

Step 6:
Click *Fill view* Button.

Step 7:
Note the maximum displacement along Y-Axis.



Results comparison

Method	Results – max displacement	Error
Analytic	3.0227mm	-
1D MES	3.0277mm	
3D MES	2.822mm	

www.agh.edu.pl

3D FEM results improvment – 1st way

Step 1:
Open model.

Step 2:
Go to *Geometry & Mesh* Tab from *Main Menu*.

Step 3:
Open Subdivide Menu.

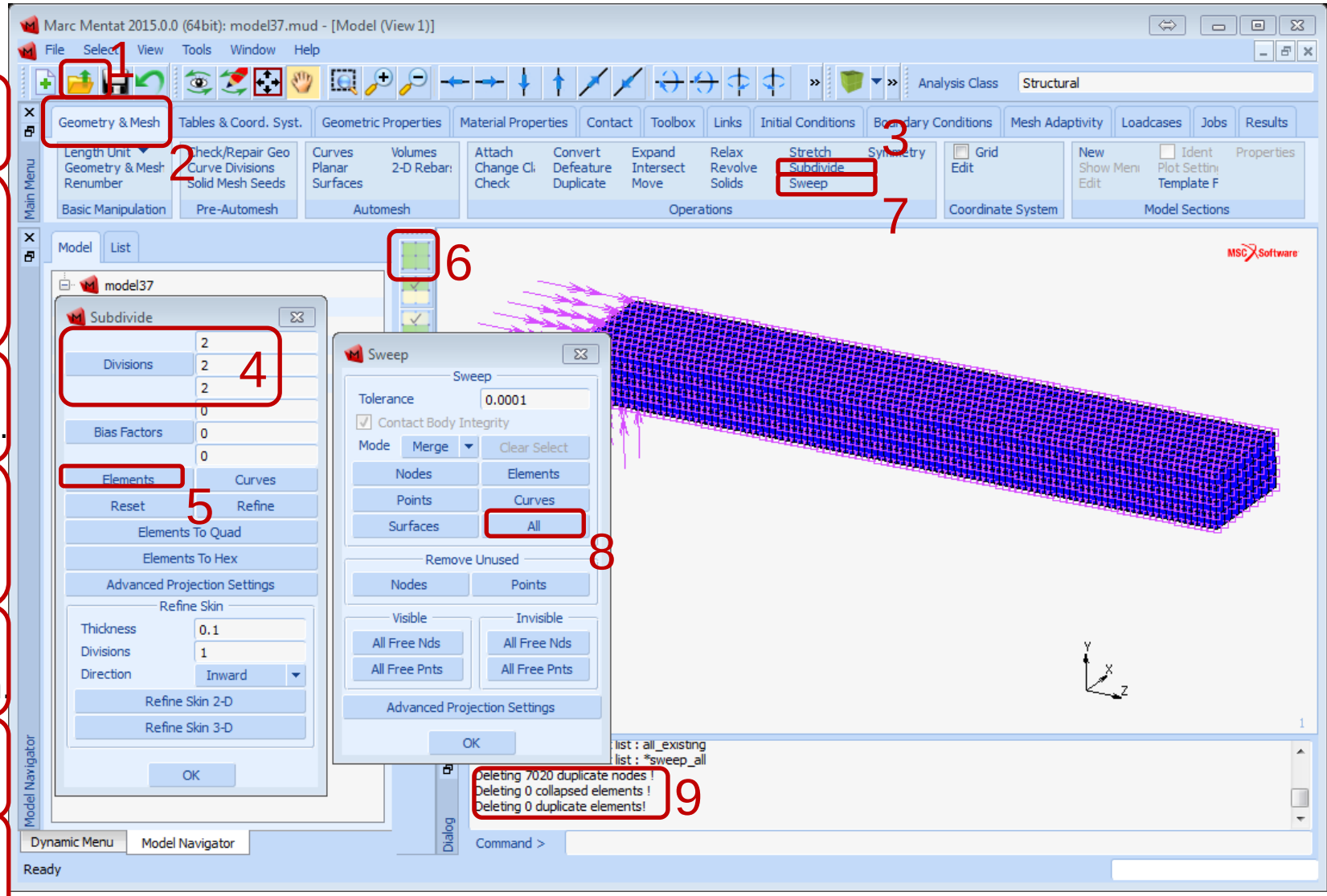
Step 4:
Divide FE 2 times along each axis.

Step 5:
Click *Elements* Button.

Step 6:
Click *all_existing* Button.

Step 7:
Open *Sweep* Menu.

Step8:
Click *All* Button – 7020 nodes should be deleted 9 .



www.agh.edu.pl

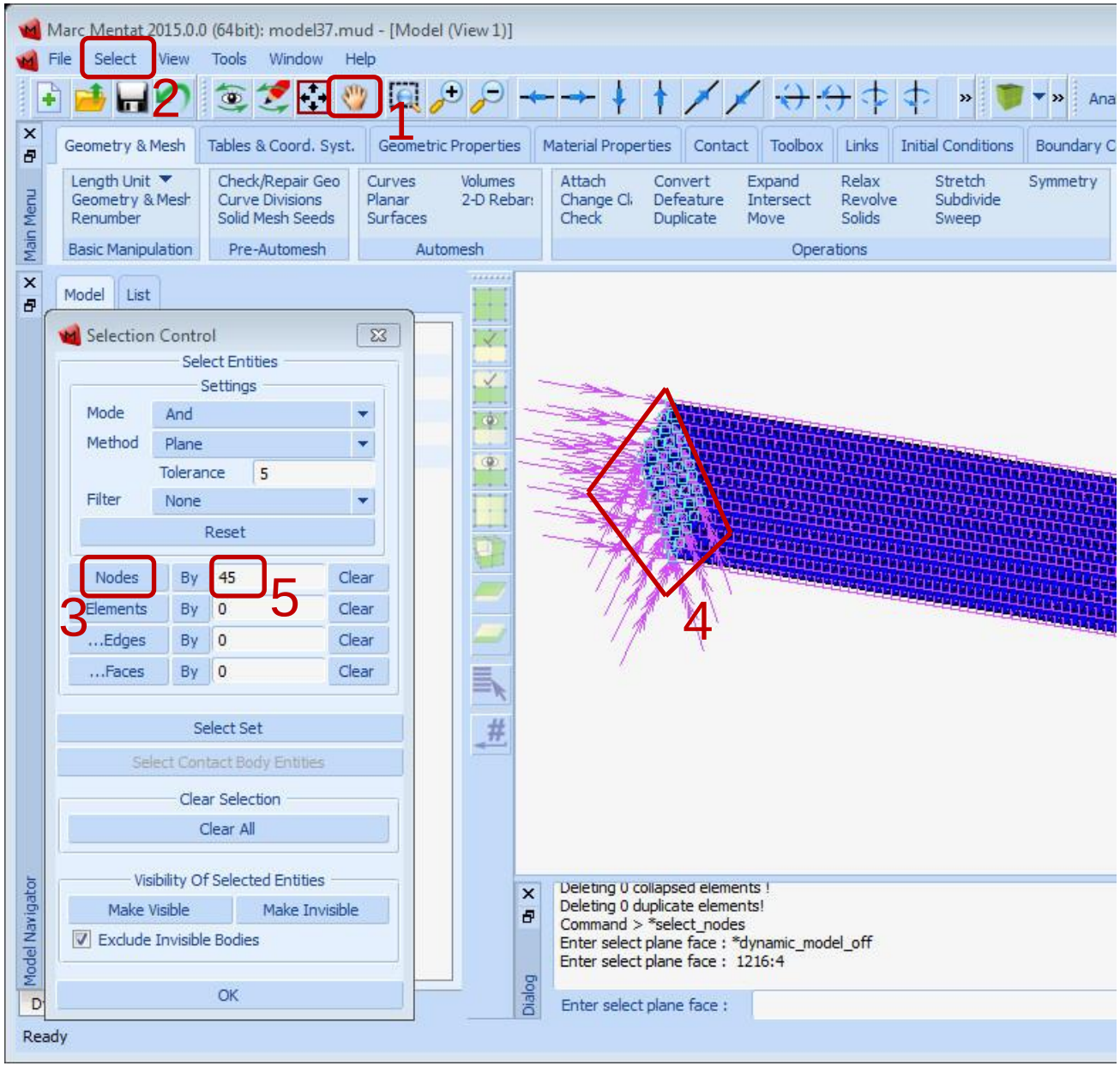
3D FEM results improvment – 1st way

Step 1:
Active *Dynamic Model* Button and rotate model.

Step 2:
From *Select* list choose *Selection Control…* option – new menu has been opened.

Step 3:
Click *Nodes* Button to choose nodes which will be fixed – deactivate *Dynamic Model* Button!

Step 4:
Pick one of nodes which lie on XY plane – after that you should note 45 selected nodes 5.



www.agh.edu.pl

3D FEM results improvment – 1st way

Step 1:
Open fixed_nodes to edit.

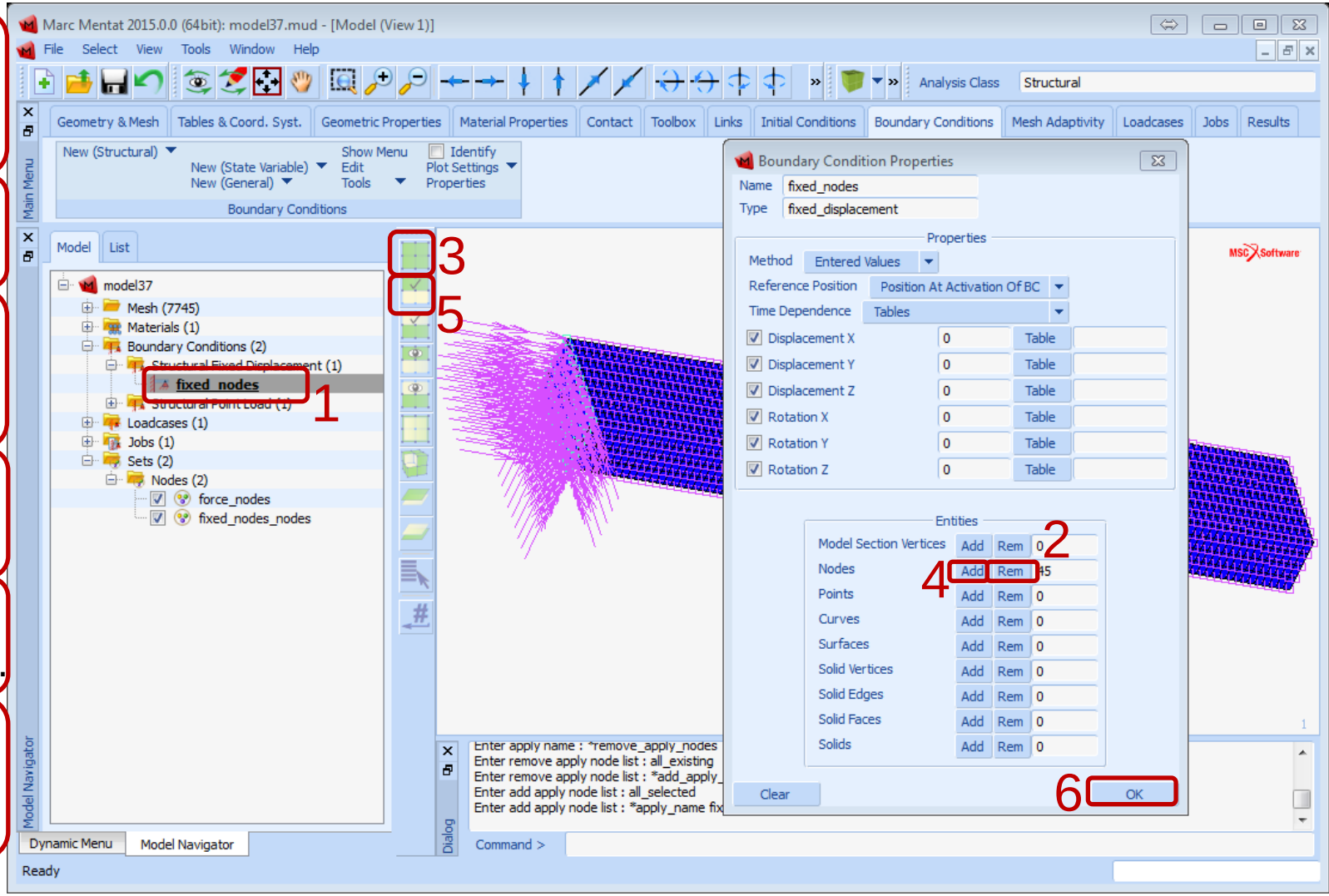
Step 2:
Click *Rem* Button.

Step 3:
Click *all_existing* Button.

Step 4:
Click *Add* Button.

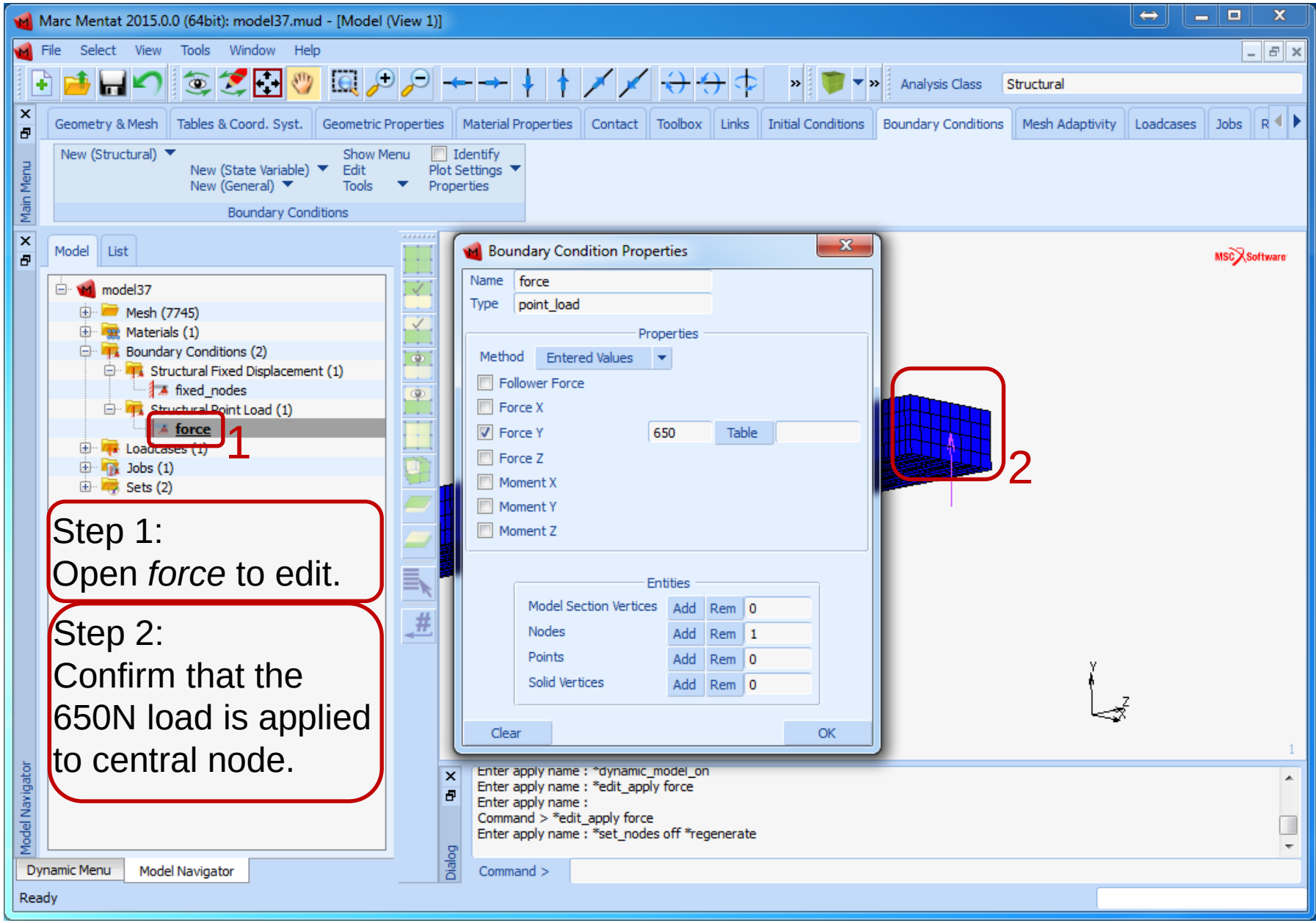
Step 5:
Click *all_selected* Button.

Step 6:
Click *OK* Button to provide changes.

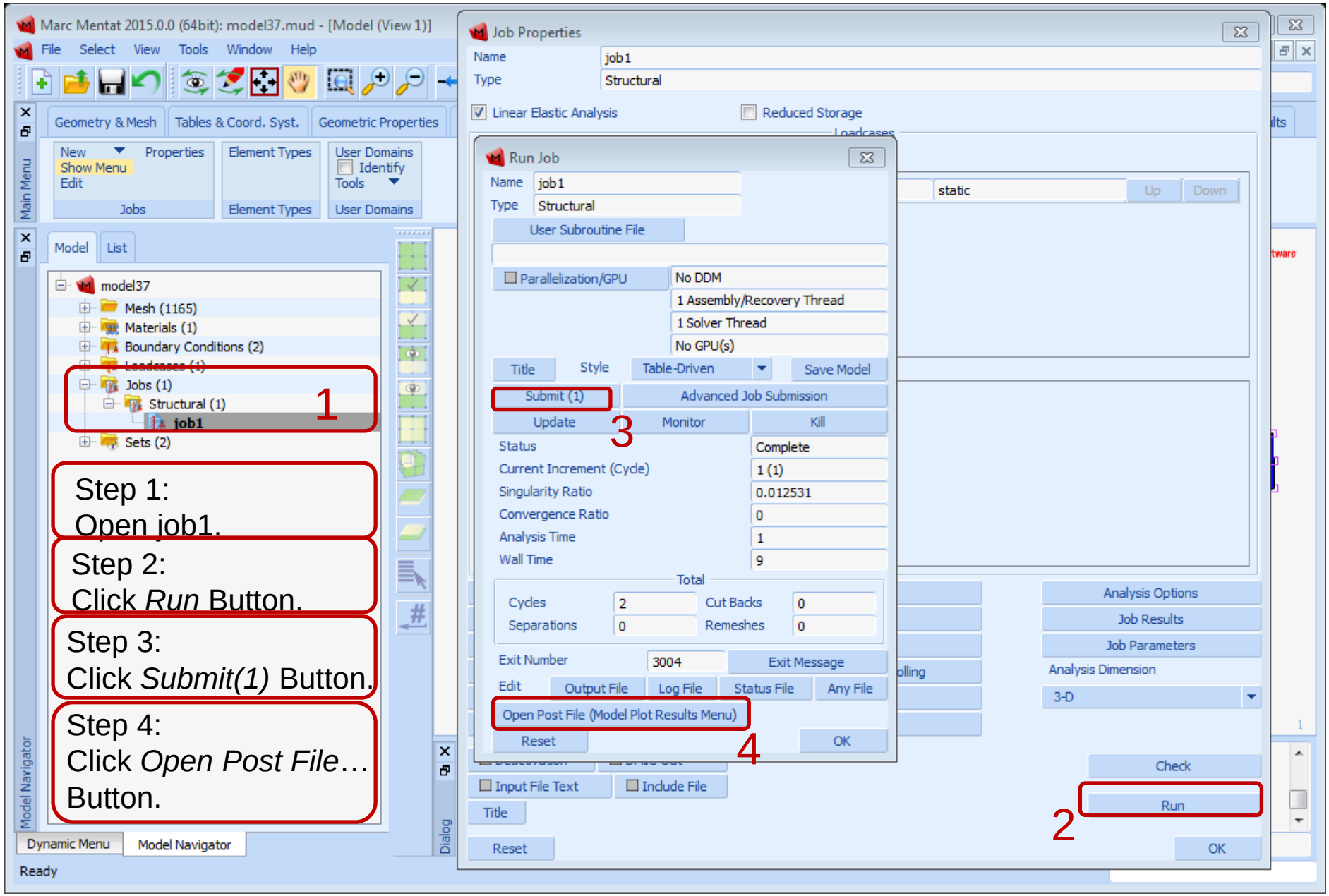


www.agh.edu.pl

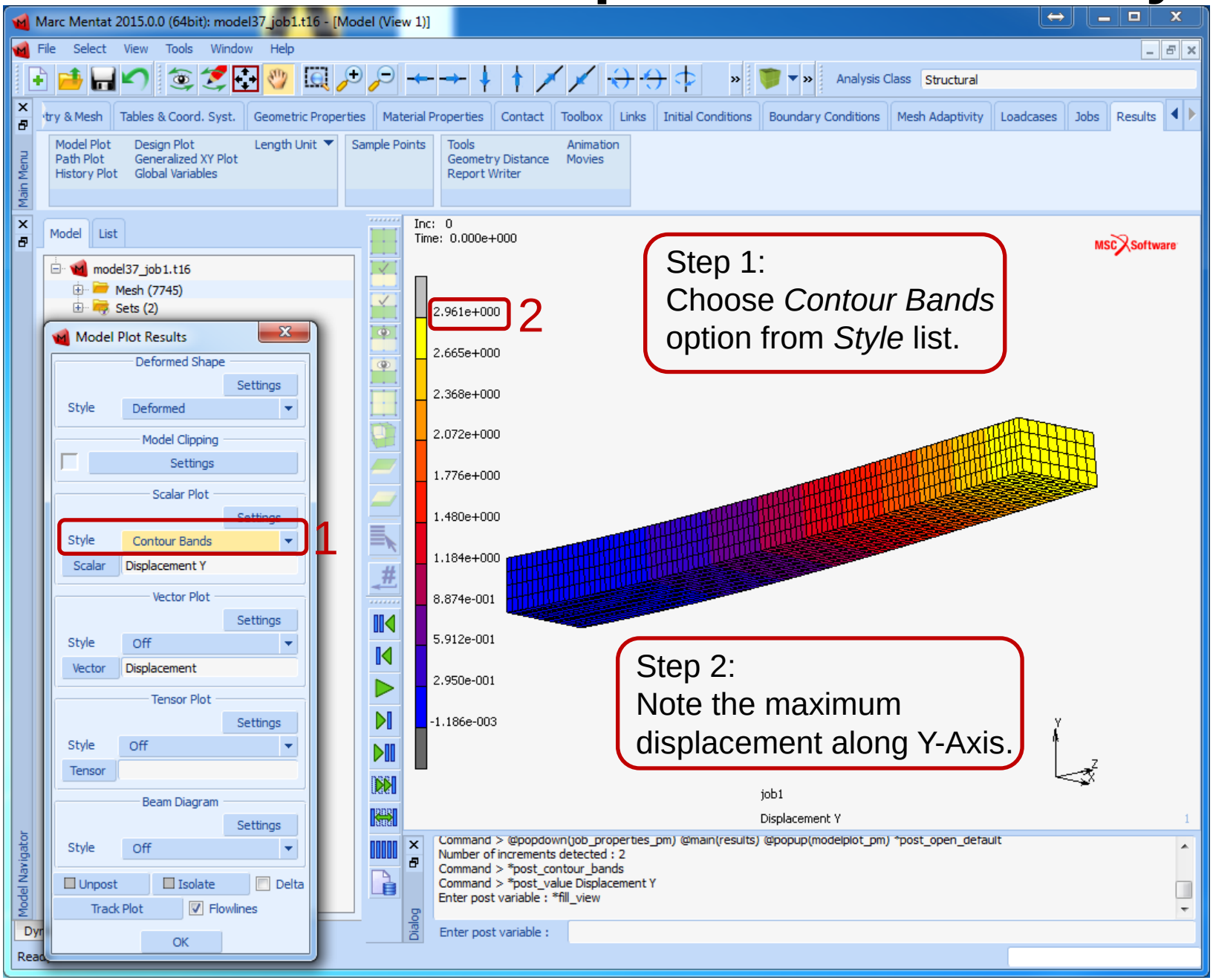
3D FEM results improvment – 1st way



3D FEM results improvment – 1nd way



3D FEM results improvment – 1st way



3D FEM results improvment – 2nd way

Step 1:
Open model.

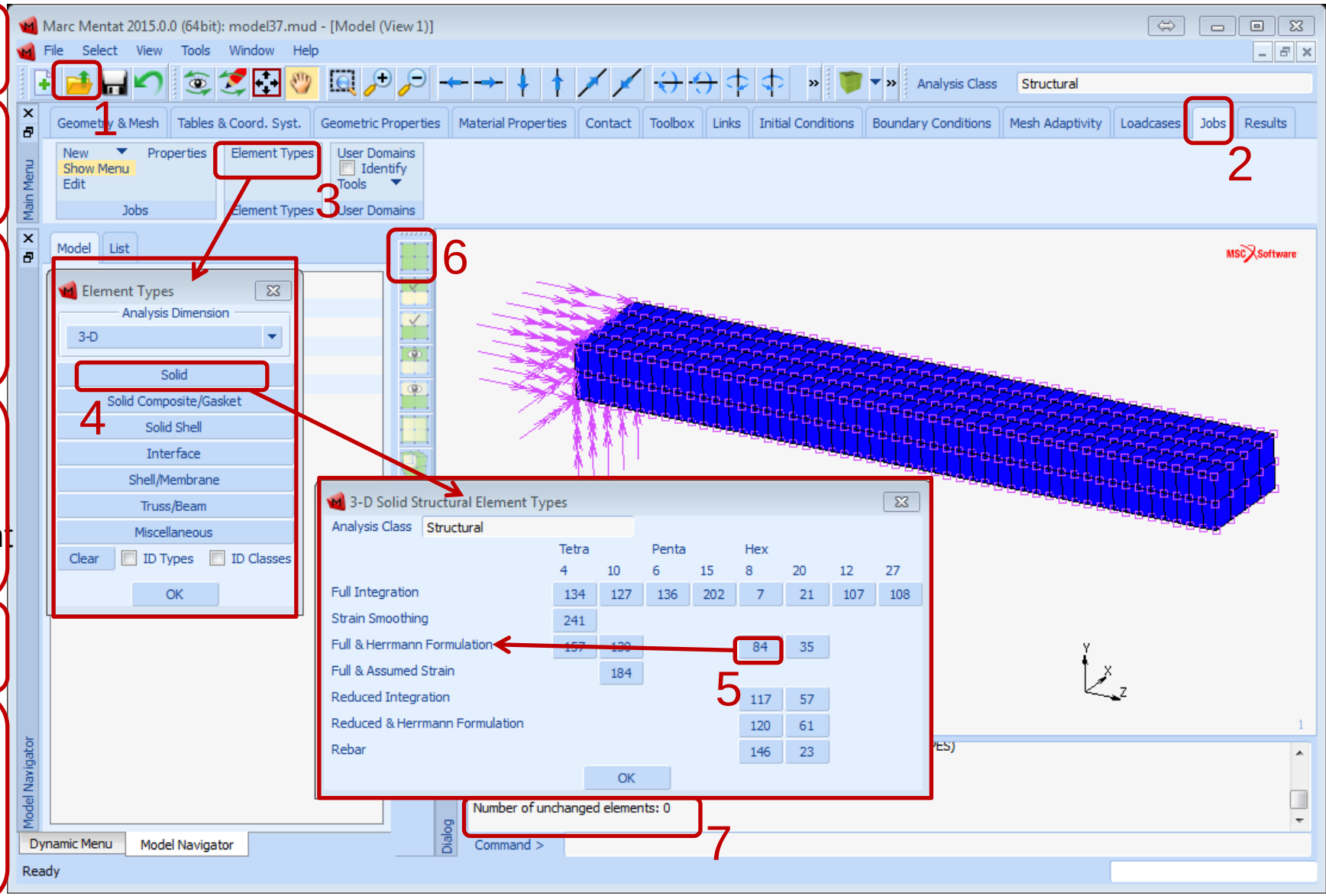
Step 2:
Go to *Jobs* Tab from *Main Menu*.

Step 3:
Click *Element Types* Button to open menu.

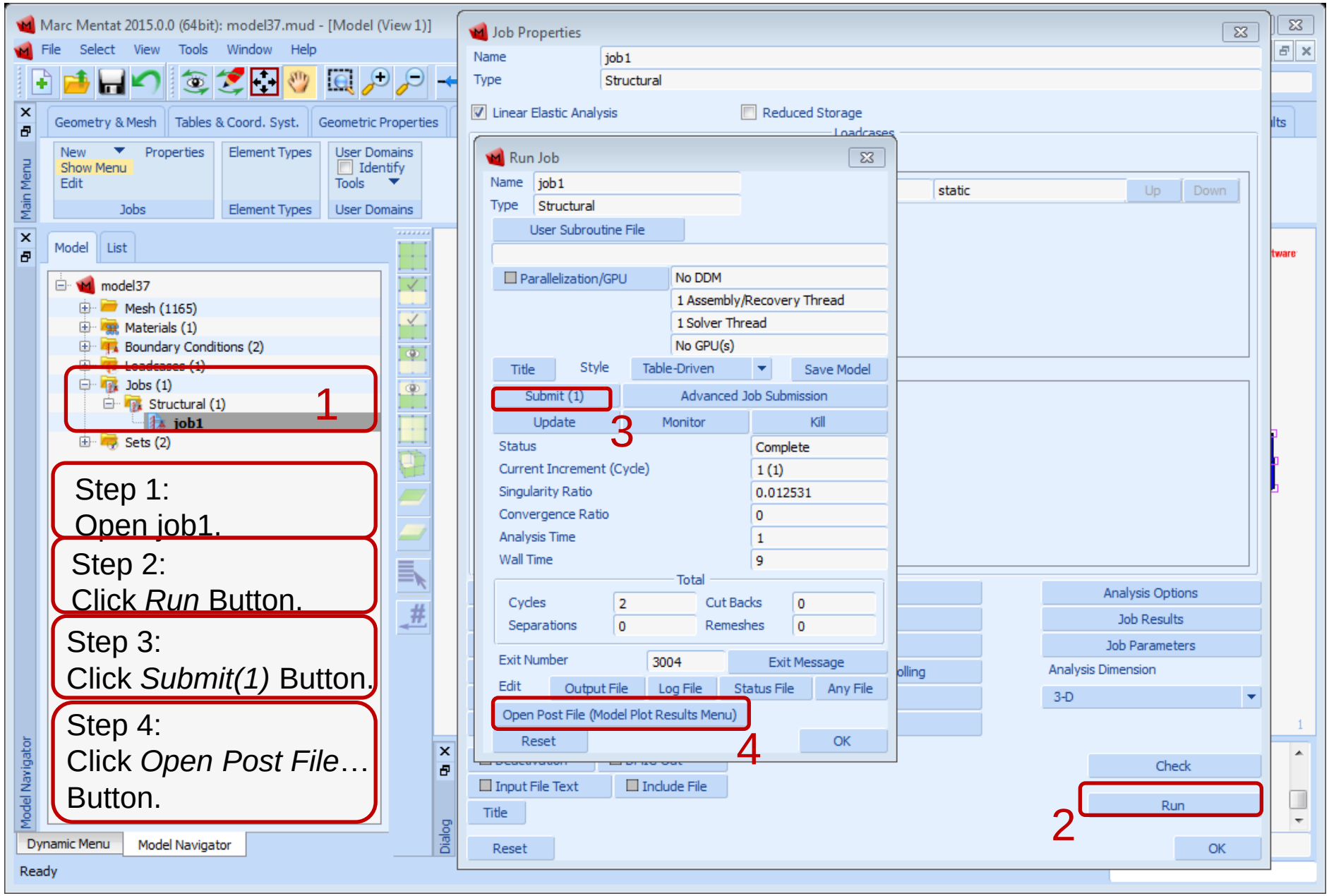
Step 4:
Click *Solid* Button to open 3-D Solid Structural Element Types menu.

Step 5:
Click *84* Button.

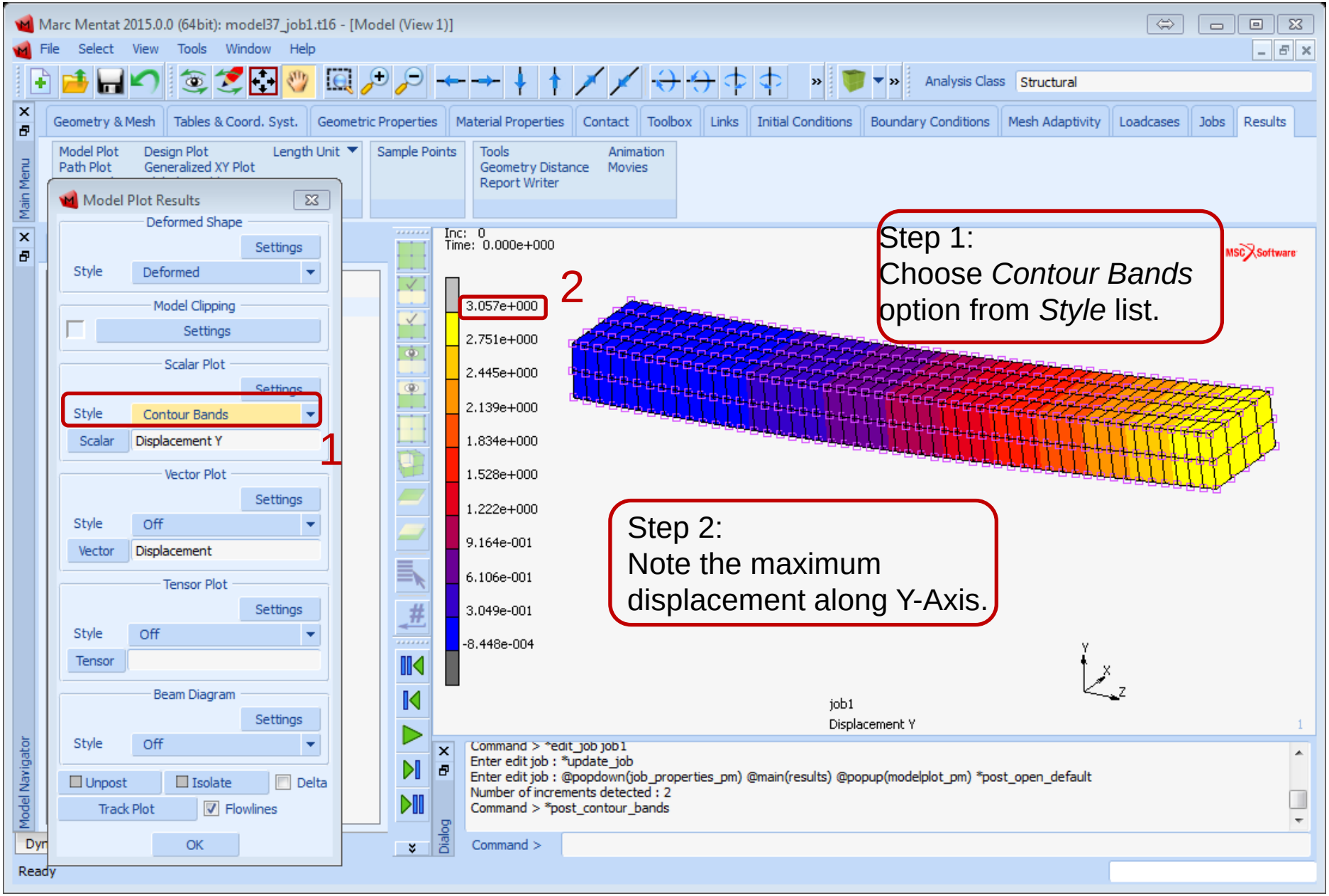
Step 6:
Click *all_exsisting* Button – none unchanged FE should left 7 .



3D FEM results improvment – 2nd way



3D FEM results improvment – 2nd way



www.agh.edu.pl

Results comparison & conclusions

Method	Number of FE	Results – max displacement	Error
Analytic	-	3.0227mm	-
1D MES	10	3.0277mm	
3D MES	400	2.822mm	
3D MES 1st improvment	3200	2.961mm	
3D MES 2st improvment	400	3.057mm	

- In 1D analysis only 10FEs were used. However, the results are better than from all 3D analysis where 400 and more Fes were used! This fact is always true for analysis where 1D and 3D FE can be used.
- The second improvement method induce better results than others 3D simulations because of appriopate fitting of FE type to stress-strain condition in the beam.
- The 1st improvment can be always done and not require the theory knowledge about FEM theory. The 2nd improvment has limitation in use and require the advance theory knowledge.